FOR EDITH · OBSTETRICS & GYNAECOLOGY JUNIOR CLERKSHIP

Diabetes Mellitus in Pregnancy

Lecture | Week 6 · Obstruction, HIV, diabetes & the written test | Obstetrics | Dr Nassimu

Placental hormones make pregnancy a state of rising insulin resistance; women who cannot compensate develop gestational diabetes, diagnosed on a 75 g OGTT (WHO 2013 fasting 5.1, 1-hour 10.0, 2-hour 8.5 mmol/L). Tight control with diet, metformin and insulin prevents macrosomia, shoulder dystocia, stillbirth and neonatal hypoglycaemia, and every woman with GDM needs lifelong screening for type 2 diabetes.

Must know before the ward round

1. Human placental lactogen, placental growth hormone, progesterone, cortisol and prolactin cause insulin resistance from the second trimester; GDM develops when the pancreas cannot double its insulin output.
2. Screen with a 75 g OGTT at 24–28 weeks after an 8–14 hour fast (earlier, and again at 24–28 weeks, if previous GDM or very high risk). Glycosuria 2+ once or 1+ twice is a trigger for an OGTT, not a diagnosis.
3. WHO 2013 75 g OGTT GDM thresholds are fasting 5.1–6.9, 1-hour 10.0 or more, 2-hour 8.5–11.0 mmol/L (any one value). Fasting 7.0 or more, or 2-hour 11.1 or more, means diabetes in pregnancy (overt).
4. HbA1c does not diagnose GDM; use it at booking to pick up overt pre-existing diabetes (48 mmol/mol, 6.5%, or more) and to judge control in known diabetes.
5. Pedersen hypothesis - maternal glucose crosses the placenta, insulin does not; the fetal pancreas makes extra insulin, which drives macrosomia, organomegaly, polyhydramnios, delayed lung maturity and neonatal hypoglycaemia.
6. Targets (NICE) are fasting below 5.3, 1-hour after meals below 7.8, or 2-hour below 6.4 mmol/L, keeping glucose above 4.0.
7. Step-up is diet and exercise, then metformin (500 mg, titrate to 2 g/day), then insulin. Start insulin at once if fasting glucose is 7.0 or more at diagnosis. Pre-existing diabetes needs folic acid 5 mg and aspirin from 12 weeks.
8. A high fasting reading needs bedtime isophane (start about 0.1–0.2 units/kg at 22:00); high post-meal readings need diet change or pre-meal soluble insulin. Titrate by about 2 units every 2–3 days.
9. Hypoglycaemia below 4.0 mmol/L is treated with 15–20 g of fast sugar and a recheck in 15 minutes; if she cannot swallow, give IV 20% dextrose 75–100 mL or glucagon 1 mg IM.
10. Deliver type 1/2 diabetes at 37+0 to 38+6 weeks; GDM on diet no later than 40+6; in labour keep capillary glucose at 4–7 mmol/L with hourly checks and an IV insulin–dextrose infusion if needed.
11. After delivery, stop all GDM treatment, feed the baby within 1 hour and check neonatal glucose at 2–4 hours (before the second feed), and test the mother at 6–13 weeks postpartum (fasting glucose or OGTT), then yearly.
12. DKA in pregnancy can occur at near-normal glucose, kills the fetus in up to a third of cases, and is treated with fluids, fixed-rate IV insulin 0.1 unit/kg/h, dextrose and potassium, not an emergency caesarean.

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1. Definition

Hyperglycaemia in pregnancy includes women who had diabetes before pregnancy (type 1 or type 2, "pre-gestational") and women whose high glucose is first detected in pregnancy, either **gestational diabetes (GDM)** or overt **diabetes in pregnancy**.

Category	Definition
Pre-existing (pre-gestational) diabetes	Type 1 or type 2 diabetes diagnosed before pregnancy
Diabetes in pregnancy (overt diabetes first detected in pregnancy)	Meets non-pregnant diabetes criteria at any gestation: fasting plasma glucose **7.0 mmol/L or more**, 2-hour OGTT **11.1 or more**, or random glucose **11.1 or more** with symptoms. Usually undiagnosed type 2 diabetes
Gestational diabetes (GDM)	Hyperglycaemia first detected in pregnancy below the overt-diabetes thresholds, typically from 24 weeks, that usually resolves after delivery

WHO 2013 criteria for a 75 g OGTT (used in Uganda)

Time	GDM (any one value)	Diabetes in pregnancy
Fasting	**5.1–6.9 mmol/L**	7.0 mmol/L or more
1 hour	**10.0 mmol/L or more**	(no 1-hour criterion)
2 hours	**8.5–11.0 mmol/L**	11.1 mmol/L or more

> NOTE
> **NICE (UK)** uses different, less strict cut-offs for GDM: **fasting 5.6 mmol/L or more, or 2-hour 7.8 mmol/L or more**. In exams, state which criteria you are using. Uganda follows **WHO 2013**.

Pregnancy is designed to be "diabetogenic": the placenta releases hormones that make the mother resistant to insulin so that more glucose goes to the baby. Most women respond by producing two to three times more insulin. Those who cannot (because of obesity, genetics or pre-existing beta-cell weakness) become hyperglycaemic.

> CLINICAL PEARL
> The one thing never to forget: **glucose crosses the placenta, insulin does not.** Every fetal complication of diabetes flows from the fetus making its own excess insulin in response to the mother's high glucose.

2. Epidemiology

- Hyperglycaemia in pregnancy affects about **1 in 6 live births worldwide**; most of it is GDM.
- In Ugandan studies using WHO 2013 criteria at 24–28 weeks (Kampala, Mbarara and elsewhere), GDM prevalence has ranged from about **7% to over 20%** (roughly 1 in 10 to 1 in 5 women screened), depending on population and screening method. Rising obesity and urbanisation drive it.

- Many facilities do not do an OGTT, and screening is selective, so **most cases are missed**. Undiagnosed GDM shows up in the labour ward as **a big baby stuck at the shoulders**, a **fresh stillbirth**, or a **floppy, jittery hypoglycaemic newborn**.
- **Pre-existing type 1 diabetes** in Uganda often comes with poor pre-pregnancy control, limited access to insulin and glucometer strips, and no pre-conception care; these pregnancies carry high rates of congenital anomalies and stillbirth.
- **Realities of facilities**: OGTT needs a fasting woman, 75 g anhydrous glucose, a 2-hour wait and a reliable glucometer or laboratory. Many HC IVs lack strips or glucose powder. Insulin is available at hospitals (human soluble, isophane/NPH and premixed 70/30), but storage, syringes and daily glucose testing at home are real barriers.
- Women with diabetes should deliver in a **hospital with a doctor able to manage shoulder dystocia, an operating theatre and a neonatal unit**.

> **UGANDA CONTEXT**
> Most Ugandan units screen **selectively** (risk factors or glycosuria) rather than universally, and many use a fasting or random capillary glucose instead of a full OGTT. When you see risk factors, **ask for an OGTT at 24–28 weeks** (earlier if very high risk). Know the WHO 2013 numbers cold; your consultants will ask.

3. Aetiology and Risk Factors

Cause. GDM is essentially **beta-cell failure unmasked by pregnancy**: chronic insulin resistance (obesity, PCOS, family history of type 2 diabetes) plus placental insulin resistance exceeds the pancreas's capacity.

Risk factors for GDM (decide who needs an OGTT)

Risk factor	Notes
BMI above 30 (Ugandan studies: BMI 25 or more)	Chronic insulin resistance
Previous GDM	Recurrence 30–50%; test early and again at 24–28 weeks
Previous baby 4 kg or more	Pattern of undiagnosed diabetes
First-degree relative with diabetes	Genetic beta-cell weakness
Previous unexplained stillbirth or neonatal death	
Polyhydramnios in this pregnancy	Fetal polyuria
PCOS	Insulin resistance
Age 35 or more (Ugandan studies: 25 or more)	
Glycosuria (2+ once or 1+ twice)	Trigger for OGTT

A substantial minority of women with GDM have **no risk factors**, so risk-factor-based screening misses cases.

Triggers of DKA in pregnancy: infection (UTI/pyelonephritis, malaria, pneumonia), vomiting/hyperemesis, missed insulin, new undiagnosed type 1 diabetes, **steroids**, **beta-agonists**, insulin pump failure.

4. Pathophysiology

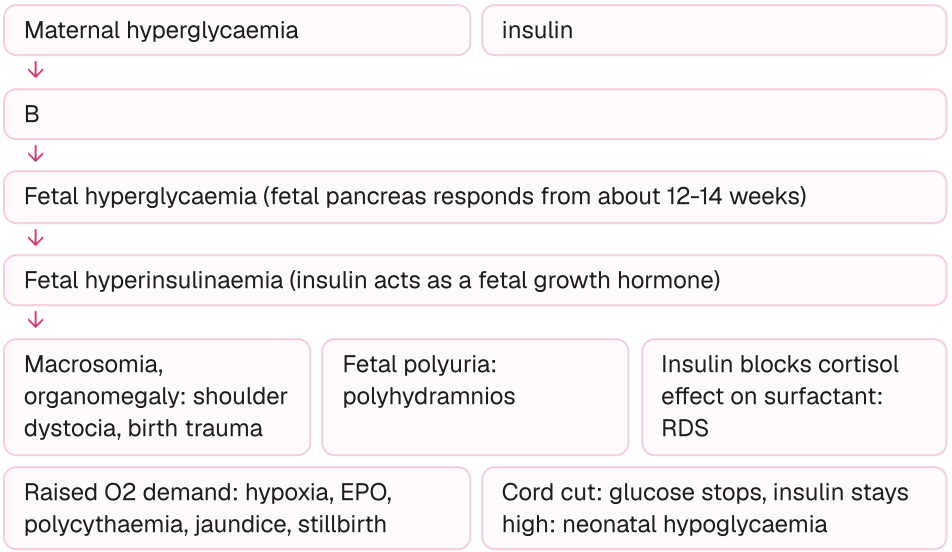
Normal glucose metabolism in pregnancy

Stage	Change	Why
First trimester	**Increased insulin sensitivity**, lower fasting glucose	Oestrogen and progesterone stimulate beta-cell hyperplasia; the fetus uses glucose continuously. **Women with type 1 diabetes have more hypoglycaemia** in early pregnancy, especially with nausea and vomiting
Second and third trimesters	**Progressive insulin resistance** (insulin sensitivity falls by about 50%); insulin secretion rises 2–3 times	**Human placental lactogen (hPL)**, placental growth hormone, progesterone, cortisol and prolactin (and placental TNF-α) antagonise insulin at the receptor/post-receptor level. The purpose is to shift glucose to the fetus while the mother burns fat
Fasting state	"**Accelerated starvation**": mother quickly switches to fat metabolism and **ketone** production	The fetus continuously drains glucose and amino acids. This is why pregnant women develop ketosis (and DKA) faster
After delivery of the placenta	Insulin resistance **vanishes within hours**	Placental hormones disappear. This is why GDM treatment is **stopped** at delivery and why women with type 1 diabetes need an immediate **cut in insulin dose**

How GDM develops

Women who develop GDM usually already had **chronic insulin resistance**. When the placenta adds its insulin resistance, their pancreas cannot keep up. This also explains why **up to half develop type 2 diabetes within 5–10 years**: the underlying defect persists.

Fetal consequences: the Pedersen hypothesis



- **Macrosomia** (birth weight 4,000 g or more, or above the 90th centile) is **asymmetric**: insulin-sensitive tissues (fat, liver, heart) grow, so the **shoulders and abdomen outgrow the head**. That is why diabetic babies get **shoulder dystocia** at lower birth weights than non-diabetic babies.
- **Stillbirth**, particularly in the last weeks, is related to fetal hyperinsulinaemia, increased oxygen demand, fetal acidaemia and placental vasculopathy. This is why we deliver women with diabetes **before 40 weeks**.
- **Neonatal hypocalcaemia and hypomagnesaemia** also occur.

Pre-existing diabetes: the first-trimester problem

High glucose around **conception and organogenesis (first 8 weeks)** is **teratogenic**. The risk of **congenital anomalies** and **miscarriage** rises with periconception HbA1c:

- **Cardiac** (transposition of the great arteries, VSD, hypertrophic cardiomyopathy)
- **Neural tube defects**
- **Caudal regression syndrome** (rare but highly specific to maternal diabetes)
- Renal agenesis, duodenal atresia

GDM, which arises after organogenesis, does **not** raise anomaly rates (unless it is actually undiagnosed type 2 diabetes).

Maternal consequences

Pre-eclampsia (endothelial dysfunction), **infections** (UTI, candidiasis, wound infection), **polyhydramnios** (preterm labour, malpresentation, cord prolapse), **worsening retinopathy and nephropathy** in pre-existing diabetes, **hypoglycaemia** (type 1), **DKA**, operative delivery and perineal trauma, and **future type 2 diabetes and cardiovascular disease**.

Why DKA is different in pregnancy

Accelerated starvation, insulin resistance, and a respiratory alkalosis with **reduced serum bicarbonate buffer** (the kidneys excrete bicarbonate to compensate for pregnancy's hyperventilation) mean DKA develops **faster** and at **lower glucose levels** (sometimes below 11 mmol/L, "euglycaemic DKA"). **Fetal loss is 10–35%.**

Feature links

- Fetal hyperinsulinaemia → macrosomia → SFH larger than dates, AC above the 90th centile → shoulder dystocia, Erb's palsy, fractured clavicle.
- Fetal polyuria → polyhydramnios → tense shiny abdomen, fetal parts hard to feel, preterm labour, malpresentation, cord prolapse.
- Insulin blocking cortisol's effect on surfactant → RDS.
- Fetal hypoxia → erythropoietin → polycythaemia → jaundice.
- Persisting fetal insulin after cord clamping → neonatal hypoglycaemia → jittery, lethargic, poor feeding, convulsions.
- Periconception hyperglycaemia → cardiac defects, NTD, caudal regression, miscarriage (pre-existing diabetes only).
- Lower renal threshold for glucose (raised GFR) → glycosuria even with normal blood glucose.
- Raised red-cell turnover → falsely low HbA1c in pregnancy.
- Maternal hyperglycaemia → polyuria, polydipsia, recurrent candidiasis and UTI (often absent in GDM).
- Accelerated starvation + low bicarbonate buffer → ketosis → vomiting, abdominal pain, Kussmaul breathing, ketotic breath, drowsiness (DKA) at near-normal glucose.
- Placental insufficiency late in pregnancy → falling insulin requirement (more than about 15–20%).
- Placental delivery → insulin resistance vanishes → maternal hypoglycaemia if GDM treatment continues.

5. Classification / Staging

Classification	Categories
By timing (WHO / current)	Pre-existing type 1; pre-existing type 2; diabetes in pregnancy (overt, first detected); GDM
By treatment of GDM (ACOG)	**A1GDM** = diet-controlled; **A2GDM** = on medication

Classification	Categories
White classification (historical)	A1 = GDM controlled by diet; A2 = GDM on medication; B to R/F by age of onset, duration and vascular complications. Still used in some textbooks and vivas
Pre-existing control (HbA1c)	Target below 48 mmol/mol (6.5%); **above 86 mmol/mol (10%)** = very high anomaly risk, advise against pregnancy
Hyperglycaemia pattern (drives treatment)	Fasting-predominant (basal insulin/metformin) vs post-meal-predominant (diet change or pre-meal soluble insulin)

6. Clinical Features

History

Two very different patients arrive with "diabetes" on the file: the **well woman referred for an abnormal glucose or a big SFH** (an outpatient diagnostic workup), and the **sick diabetic woman** who is vomiting, drowsy or hypoglycaemic (an emergency, section 9).

Ask	Why it matters
Type of diabetes and when diagnosed: known type 1 or 2, GDM this pregnancy, or "sugar found in the urine"? Duration?	Pre-existing diabetes brings anomaly, stillbirth and DKA risk
Current treatment, exact doses and timing: metformin dose, insulin type (soluble, isophane, 30/70), units and times	You will need these to titrate; oral agents other than metformin must be switched
Home glucose readings (ask to see the book or the meter); last HbA1c: which readings are high, fasting or post-meal?	Control; decides which dose to change
Hypoglycaemia: how often, how low, does she feel it coming?	Recurrent hypos or unawareness need a dose cut; type 1 women lose hypo awareness in pregnancy
Gestational age by the best dating	Decides OGTT timing, delivery timing and steroid use
Pre-conception care: folic acid 5 mg? When started? **Aspirin**?	Neural tube defect risk; pre-eclampsia prevention
Known complications (pre-existing): eyes, kidneys, BP, neuropathy, heart disease	Retinopathy can progress rapidly; nephropathy predicts pre-eclampsia and FGR
Medicines: ACE inhibitors, ARBs, statins (stop), sulphonylureas and other oral agents (switch), **steroids given recently** (glucose rises for 3–5 days), **TLD** (dolutegravir raises metformin levels)	Safety and dosing
GDM risk factors (if not yet diagnosed): section 3	Decide who needs an OGTT
Polyuria, polydipsia, recurrent candidiasis or UTIs, blurred vision	Symptoms of hyperglycaemia (often absent in GDM)
Previous pregnancies: birth weights, shoulder dystocia, stillbirth, neonatal hypoglycaemia or admission, congenital anomalies	Pattern of undiagnosed diabetes; guides delivery planning and counselling
Vomiting, abdominal pain, breathlessness, drowsiness, fever, dysuria	DKA and its triggers
Fetal movements	Reduced movements precede stillbirth
Diet, activity, occupation, money for food and strips, insulin storage	The plan must fit her life

Examination

Follow the order of Hutchison's Clinical Methods (24th edition), with the obstetric abdominal examination in its standard sequence. In a sick woman, capillary glucose and ABCDE come first (section 9).

Introduction, consent and preparation

- Introduce yourself; confirm her name, age and gestation; explain and gain consent; offer a chaperone.
- **Hand hygiene**.
- **Position and exposure**: semi-recumbent with a left lateral tilt; expose the abdomen; injection sites on the abdomen and thighs.

General inspection and vital signs

Area	Look for	Document
General	**BMI**, hydration, pallor, **Kussmaul breathing**, ketotic (fruity) breath, drowsiness/GCS	"BMI 34, well hydrated, not acidotic"
Vital signs	**BP** (pre-eclampsia), pulse (tachycardia in DKA), temperature (infection trigger), RR, hypotension	"BP 124/78, afebrile, RR 16"
Bedside	Capillary glucose; **urine dipstick** (glucose, ketones, protein, nitrites)	

Hands, face and neck

- **Hands**: capillary refill and cool peripheries (dehydration in DKA); finger-prick sites.
- **Face**: dry mucous membranes, sunken eyes (DKA); **fundoscopy** if you can (pre-existing diabetes; refer for a dilated retinal exam).
- **Neck**: thyroid (autoimmune thyroid disease in type 1).

Injection sites, feet and other systems

- **Injection sites**: lipohypertrophy causes erratic absorption. "No lipohypertrophy on the abdomen or thighs."
- **Feet** (pre-existing): sensation, peripheral neuropathy, foot ulcers. "Retinal screen booked; feet intact."
- **Signs of infection**: loin tenderness (pyelonephritis), vulvovaginal candidiasis, wound or chest signs. Infection is the commonest DKA trigger.
- **Oedema** (pre-eclampsia, nephropathy).

Obstetric abdominal examination

Step	Finding	What it means
Inspection	Distension; tense, shiny abdomen	Polyhydramnios
Fundal height	**SFH larger than dates** (more than 2 cm, some use 3 cm, above gestation in weeks)	Macrosomia or polyhydramnios: scan
Lie and **presentation**	Malpresentation	Polyhydramnios
Engagement (fifths)	High head at term	Macrosomia, disproportion
Palpation	Fetal parts hard to feel	Polyhydramnios
Auscultation of the fetal heart	Hard to hear	Polyhydramnios

In labour: clinical EFW, pelvic assessment, progress on the partograph (slow late first stage with a big baby = shoulder dystocia risk).

Completing the examination

- Thank her, cover her, **wash hands**, **summarise**.

Document: "SFH 36 cm at 32 weeks, liquor clinically increased, cephalic, 5/5 palpable, FHR 146".

Red flags

RED FLAG

- **Vomiting, abdominal pain, rapid deep breathing, drowsiness in a woman with diabetes**: DKA until proven otherwise, even if glucose is only mildly raised.
- **Recurrent hypoglycaemia** (below 4.0 mmol/L) or hypoglycaemia unawareness in type 1 diabetes.
- **Reduced fetal movements**: CTG/ultrasound the same day; stillbirth risk is high.
- **Rising BP or proteinuria**: pre-eclampsia.
- **Rapidly increasing SFH or polyhydramnios**: poor control, preterm labour risk.
- **A falling insulin requirement in the third trimester** (more than about 15–20%): may signal **placental insufficiency**; assess the fetus.
- **Suspected macrosomia in labour with slow progress in the late first stage or a prolonged second stage**: prepare for shoulder dystocia.

Thinking like a clinician

- **Glycosuria in pregnancy is not diagnostic**: the renal threshold for glucose falls in pregnancy (increased GFR), so many normal women show glycosuria. But **glucose 2+ once or 1+ twice** should prompt an OGTT.
- **HbA1c is not used to diagnose GDM** (it falls in pregnancy due to increased red-cell turnover, and it is affected by anaemia and haemoglobinopathies common in Uganda). It is useful at booking to detect overt pre-existing diabetes and to judge control in pre-existing diabetes.
- **A large-for-dates uterus has a differential**: wrong dates, macrosomia, polyhydramnios, multiple pregnancy, fibroids, molar pregnancy. Diabetes is one cause, not the only one; confirm with ultrasound.
- **Polyhydramnios should always make you think of diabetes** (and of fetal anomalies such as oesophageal atresia).
- **Fasting hyperglycaemia and post-meal hyperglycaemia are treated differently**: a high fasting value needs night-time basal insulin (isophane) or metformin; high post-meal values need diet adjustment or pre-meal short-acting insulin.
- **Steroids for fetal lung maturity will push glucose up** for about 3–5 days: increase insulin and monitor closely; do not withhold steroids because of diabetes.
- **Avoid beta-agonist tocolysis (salbutamol, terbutaline)** in diabetics: they cause hyperglycaemia and can precipitate DKA. Nifedipine is preferred.
- **Common student mistakes**:
  1. Continuing insulin/metformin after delivery in a woman with GDM (causes maternal hypoglycaemia).
  2. Forgetting the baby: early feeding and glucose checks are part of the mother's management plan.
  3. Using a random glucose to rule out GDM.
  4. Treating DKA in pregnancy by rushing to caesarean section for an abnormal CTG. **Fix the mother first**; the CTG usually improves.
  5. Forgetting the **6–13-week postpartum test**.

7. Differential Diagnoses

Differential	Distinguishing clinical feature	Distinguishing investigation
GDM	Usually asymptomatic; detected on OGTT at 24–28 weeks; resolves after birth	75 g OGTT (WHO 2013)
Overt (pre-existing undiagnosed) type 2 diabetes	Fasting 7.0 or more, HbA1c 6.5% (48 mmol/mol) or more at booking; may have complications; persists postpartum	Fasting glucose, HbA1c at booking; postpartum OGTT
Type 1 diabetes	Young, lean, known insulin dependence, prone to DKA	Known history; ketones
Physiological glycosuria of pregnancy	Glycosuria with normal blood glucose	Normal OGTT
Steroid-induced hyperglycaemia	After betamethasone/dexamethasone doses	Timing; resolves in days
Stress hyperglycaemia (sepsis, severe malaria)	Acutely unwell; resolves with treatment	Repeat testing when well
Large-for-dates from other causes	Wrong dates, twins, fibroids, polyhydramnios from anomalies	Ultrasound
DKA vs other causes of vomiting and abdominal pain (hyperemesis, pyelonephritis, appendicitis, abruption, pre-eclampsia/HELLP)	DKA: ketonuria 2+ or more, acidosis, Kussmaul breathing, glucose may be only mildly raised	Blood/urine ketones, blood gas or bicarbonate, glucose

8. Clinical Workup

Bedside

Test	How / when	What the result means	Available at
Capillary blood glucose (glucometer)	On arrival; then per regimen; **hourly in labour**	Below 4.0 hypoglycaemia; in labour target **4–7 mmol/L**	HC III and above (strips permitting)
Capillary glucose profile	Fasting + **1 hour after each main meal** (GDM on diet/metformin); add pre-meal and bedtime readings on insulin	Tells you *which* dose to change (worked example 2)	Home glucometer where possible
Urine dipstick	Every visit	**Glucose** 2+ once or 1+ twice → OGTT; **ketones** 2+ or more when unwell → think DKA; **protein** → pre-eclampsia; **nitrites/leucocytes** → UTI	All levels
Blood ketone meter (where available)	Glucose above 11 or unwell	3.0 mmol/L or more with acidosis = DKA	Few units
BP	Every visit	Pre-eclampsia is commoner	All levels
SFH tape measure	Every visit from 20 weeks	SFH more than 2 cm (some use 3 cm) above gestation in weeks: scan for macrosomia/polyhydramnios	All levels
Fetal heart (Pinard/Doppler) and CTG	Every visit; CTG with reduced movements, in labour (continuous for pre-existing and insulin-treated)	Fetal wellbeing	CTG at hospitals

Test	How / when	What the result means	Available at
Point-of-care ultrasound	If SFH large or presentation unclear	Deepest pool above 8 cm or AFI above 24–25 cm = polyhydramnios	HC IV and above
mRDT, HemoCue	Fever / pallor	Malaria is a DKA trigger; anaemia lowers HbA1c	All levels

> NOTE
> A glucometer reading is **capillary whole blood**; the OGTT criteria are for **venous plasma**. A glucometer is acceptable for monitoring, and for the OGTT only if the laboratory cannot do it, but it is less accurate at diagnosis. Use a calibrated meter and in-date strips.

Basic lab

Test	When	Expected / abnormal finding	How it changes management	Available at
75 g OGTT (venous plasma, fluoride tube)	**24–28 weeks**; at booking/early if previous GDM or very high risk (repeat at 24–28 weeks if normal)	WHO 2013 GDM: **fasting 5.1–6.9, 1 h 10.0 or more, 2 h 8.5–11.0** (any one); overt: **fasting 7.0 or more or 2 h 11.1 or more**	Diagnosis; overt diabetes or fasting 7.0 or more → **insulin at once**	Possible at many HC IVs (glucose powder, strips, 2-hour wait); reliable at hospitals
Fasting plasma glucose ± HbA1c at booking (high-risk women)	First visit	FPG 7.0 or more or HbA1c **48 mmol/mol (6.5%) or more** = overt diabetes	Treat as pre-existing diabetes: early anomaly scan, joint clinic	FPG at HC IV; HbA1c at referral labs
Urine culture	Booking, symptoms, positive nitrites	Asymptomatic bacteriuria (commoner in diabetes)	Treat; prevents pyelonephritis and DKA triggers	Referral labs
Postpartum fasting glucose or 75 g OGTT (after GDM)	**6–13 weeks** (OGTT at 4–12 weeks per WHO/ACOG)	Fasting 6.0–6.9 high risk; 7.0 or more likely type 2	Lifestyle programme, confirm and treat diabetes, annual testing	HC IV and above
Neonatal glucose	2–4 hours after birth, before the second feed, then before feeds for 24 hours	Below local threshold (2.6 mmol/L in many Ugandan units; NICE 2.0)	Feed support; IV 10% dextrose 2 mL/kg if symptomatic	All delivering facilities with a glucometer

How to do an OGTT properly

1. Normal carbohydrate diet for 3 days beforehand.
2. **Overnight fast of 8–14 hours** (water allowed).
3. Measure fasting plasma glucose (venous sample in a fluoride tube, or a calibrated glucometer if that is all you have).
4. Give **75 g anhydrous glucose dissolved in 250–300 mL of water**, drunk within 5 minutes.
5. Sit quietly (no walking about, no food, no smoking).
6. Measure glucose at **1 hour and 2 hours**.
7. Do not do the test if she is acutely unwell, vomiting, or within about a week of antenatal corticosteroids (glucose is falsely raised).

Imaging

Test	When	Expected / abnormal finding	How it changes management	Available at
Dating ultrasound	First visit / early	Accurate GA	Timing of OGTT and delivery	HC IV and above
Detailed anomaly scan with cardiac views (pre-existing or overt)	**18–22 weeks**	Cardiac defects, NTD, caudal regression	Counselling; delivery at a centre with paediatric cardiology/surgery	Regional/national referral
Growth scans	Pre-existing: every 4 weeks **28–36 weeks**; GDM: from diagnosis (e.g. 28, 32, 36 weeks)	**AC above the 90th centile**, EFW 4,000 g or more, polyhydramnios; FGR in vasculopathy	AC above the 90th centile = tighten control (consider insulin); EFW 4,500 g or more → offer CS; FGR → closer surveillance. EFW is inaccurate (about ±15%) in macrosomia	Hospitals
CTG	Reduced movements, from about 36–38 weeks per protocol, continuous in labour	Non-reassuring features	Delivery decisions (but correct DKA first)	Hospitals

Specialised / referral-level

Test	When	Expected / abnormal finding	How it changes management	Available at
HbA1c (pre-existing diabetes)	Booking, then each trimester	Target below 48 mmol/mol; above 86 mmol/mol (10%) = very high anomaly risk	Intensify treatment; detailed anomaly scan. Falsely low with anaemia, recent transfusion, sickle cell	Regional/national referral or private laboratories
Creatinine, electrolytes, urine albumin:creatinine ratio (pre-existing)	Booking	Raised creatinine, albuminuria = nephropathy	Predicts pre-eclampsia and FGR; drug dosing; metformin contraindicated if eGFR below 30	Hospitals
Retinal assessment (pre-existing)	Booking and 28 weeks	Retinopathy (can progress as control tightens)	Ophthalmology referral; timing of tightening control	Referral hospitals
Thyroid function (type 1)	Booking	Autoimmune thyroid disease	Treat	Referral hospitals
DKA bundle: venous/capillary ketones, **blood gas or serum bicarbonate**, **K^+**, sodium, creatinine, FBC, blood/urine cultures, malaria test	Any suspicion of DKA	Ketones 3.0 or more (or urine 2+ or more) **plus** bicarbonate below 15 or pH below 7.3	Confirms DKA; **K^+ guides potassium replacement**; finds the trigger	Blood gas only at a few referral units: use serum bicarbonate and clinical Kussmaul breathing where it is lacking

Putting it together

Model one-line summary

> "A 32-year-old G3P2 at 29 weeks, BMI 33, with a previous 4.2 kg baby, diagnosed with GDM on a 75 g OGTT at 26 weeks (fasting 5.6, 1 h 10.2, 2 h 8.1 mmol/L), on diet for 2 weeks with persistently raised fasting

> readings, BP normal, SFH 31 cm, no ketonuria."

Problem list

1. GDM (WHO 2013), not at target on diet alone, **fasting-predominant** hyperglycaemia.
2. Previous macrosomic baby: risk of macrosomia and **shoulder dystocia**.
3. Obesity (BMI 33): pre-eclampsia and thromboembolism risk.
4. Plan for delivery timing, neonatal glucose checks and postpartum testing.

Working diagnosis: gestational diabetes mellitus requiring pharmacological treatment.

Model case presentation (ward round)

> "Mrs K is a 34-year-old G4P3+0 at 30 weeks, BMI 34, whose previous baby weighed 4.3 kg with a difficult delivery of the shoulders. Her 75 g OGTT at 26 weeks showed **fasting 5.8, 1-hour 10.9 and 2-hour 9.1 mmol/L**, meeting WHO 2013 criteria for **GDM**. After 2 weeks of diet her fasting readings remain 5.6–6.0 with post-meal values of 7.0–8.5, so she was started on **metformin 500 mg twice daily**, now 1 g twice daily. BP is 124/78, urine clear. SFH is 33 cm; a scan at 28 weeks showed an **AC above the 90th centile** with normal liquor. **Plan**: continue metformin, add **bedtime isophane** if fasting stays above 5.3, growth scan at 32 and 36 weeks, fetal movement counting, plan **induction around 38–39 weeks** with a senior present for possible shoulder dystocia, hourly glucose in labour, stop metformin after birth, neonatal glucose checks, and **fasting glucose/OGTT at 6–13 weeks postpartum**."

Worked example 1: reading a 75 g OGTT (WHO 2013)

Woman	Fasting	1 h	2 h	Interpretation
A	4.6	**10.4**	8.0	**GDM** on the 1-hour value alone (any one value is enough)
B	**5.3**	9.4	7.4	**GDM** by WHO 2013 (fasting 5.1 or more). By NICE (fasting 5.6, 2 h 7.8) she would **not** have GDM: state which criteria you use
C	**7.4**	13.0	**12.2**	**Diabetes in pregnancy (overt)**: fasting 7.0 or more and 2 h 11.1 or more. Start **insulin at once**, early detailed scan, check HbA1c, retest postpartum
D	4.8	9.6	8.3	**Normal** (every value below threshold). If very high risk and tested early, repeat at 24–28 weeks

EXAM FAVOURITE

Common exam mistake: calling woman D "borderline GDM". There is no borderline in WHO 2013: **each value is compared with its own cut-off (5.1 / 10.0 / 8.5)** and **one** abnormal value makes the diagnosis.

Worked example 2: a capillary glucose profile on metformin. Woman on **metformin 1 g twice daily** for 3 weeks, weight 80 kg. Three-day profile (mmol/L):

Day	Fasting	1 h after breakfast	1 h after lunch	1 h after supper
1	**6.1**	7.2	7.5	7.4
2	**6.0**	7.0	7.6	7.1
3	**5.9**	7.3	7.2	7.5

Reading it: all post-meal values are below 7.8 (on target); **every fasting value is above 5.3**. The problem is overnight hepatic glucose output, not meals.

Action: metformin is at 2 g/day (the usual maximum). **Add bedtime isophane (NPH) at 22:00**, starting at about **0.1–0.2 units/kg** (8–16 units for 80 kg; start at the lower end, e.g. **8 units**), and increase by **2 units every 2–3 days** until fasting is below 5.3 mmol/L, without any reading below 4.0. Teach hypoglycaemia recognition and treatment. If post-meal values later rise, add **pre-meal soluble insulin** to the meal that is high.

Worked example 3 (quick interpretation): is this DKA? Glucose 13.2 mmol/L, urine ketones 3+, serum bicarbonate 12 mmol/L, RR 32 with deep breaths, vomiting for 2 days, febrile with loin pain. **Yes: DKA** (ketosis plus acidosis) with a probable **pyelonephritis trigger**. Fluids, fixed-rate insulin 0.1 unit/kg/h, potassium guided by K^+, dextrose when glucose falls below 14, IV antibiotics, fetal monitoring once stable. **No emergency CS for the CTG.**

9. Management

Immediate / emergency

First 5 minutes (any woman with known or suspected diabetes who is unwell, in labour or newly admitted)

1. **Capillary glucose now.**
   - **Below 4.0 mmol/L**: treat at once. Conscious: **15–20 g fast sugar** (3–4 teaspoons of sugar in water, or 150–200 mL of soda), recheck in 15 minutes, then a snack. Unconscious or unable to swallow: **IV 75–100 mL of 20% dextrose** (or 150–200 mL of 10%; 50% dextrose 25–50 mL only if nothing else, via a large vein, as it is irritant) or **glucagon 1 mg IM**.
   - **Above 11 mmol/L, or unwell with vomiting**: check **ketones** (urine dipstick or a blood ketone meter).
2. **ABCDE**: pulse, BP, RR (deep sighing **Kussmaul breathing** = acidosis), temperature, SpO_2, GCS, signs of dehydration. Left lateral tilt.
3. **Suspect DKA** if she is vomiting, has abdominal pain, is breathing deeply, is drowsy, or has **urine ketones 2+ or more**, even with glucose only mildly raised. **Call the obstetrician, physician and anaesthetist**, put in two large-bore cannulae, start **0.9% sodium chloride**, and follow the DKA algorithm below.
4. **Look for the trigger**: fever (mRDT, urine), vomiting, missed insulin, recent **steroids** or **salbutamol**.
5. **Fetal heart**: listen, but remember that an abnormal CTG in DKA usually recovers when the mother is corrected. **Stabilise the mother first.**
6. **In labour**: start **hourly capillary glucose** (target 4–7 mmol/L) and alert the neonatal team.

DKA in pregnancy: diagnosis and algorithm

Diagnosis: ketonaemia (blood ketones 3.0 mmol/L or more, or urine ketones 2+ or more) **plus** acidosis (bicarbonate below 15 mmol/L or venous pH below 7.3), with raised glucose or known diabetes.

1. **Call for help**: obstetrician, physician, anaesthetist; HDU/ICU if available. **ABCDE**. Left lateral tilt.
2. **Two large-bore IV cannulae**; bloods: glucose, ketones, electrolytes (especially **K^+**), creatinine, FBC, blood gas/bicarbonate, cultures, malaria test; urinalysis.
3. **Fluids**: **0.9% sodium chloride**. If SBP below 90 mmHg: 500 mL over 10–15 minutes, repeat. Otherwise, for example, **1 L over 1 hour, 1 L over 2 hours, 1 L over 2 hours, 1 L over 4 hours, 1 L over 4 hours**, then reassess. Be more cautious if pre-eclampsia or cardiac/renal disease (risk of pulmonary oedema).
4. **Fixed-rate IV insulin**: **soluble insulin 0.1 unit/kg/hour** (weight at presentation) by syringe driver. Continue long-acting insulin if she normally takes it.
   - If no syringe driver: follow the UCG/ward protocol for hourly IM/SC soluble insulin; do not delay insulin.
5. **Potassium** (from the second litre, guided by K^+):
   - K^+ above 5.5 mmol/L: none.
   - K^+ 3.5–5.5 mmol/L: **40 mmol KCl per litre**.

- K^+ below 3.5 mmol/L: senior review; more potassium via a central/large line; may need to delay insulin until K^+ rises.

6. **Add 10% dextrose at 125 mL/hour** alongside saline **when glucose falls below 14 mmol/L**, continuing insulin until ketones clear (the insulin is needed to switch off ketogenesis, not just to lower glucose).
7. **Monitor**: glucose and ketones hourly, K^+ and bicarbonate at 1, 2 and 4 hours then 4-hourly, fluid balance with a urinary catheter, GCS.
   - Aims: ketones fall by at least 0.5 mmol/L/hour, bicarbonate rises by 3 mmol/L/hour, glucose falls by 3 mmol/L/hour.
8. **Treat the trigger** (antibiotics for infection, antimalarials).
9. **Fetal monitoring** once the mother is stable. **Abnormal CTG in DKA reflects maternal acidosis and usually recovers as she is corrected: do not perform an emergency caesarean on an acidotic, unstable mother.**
10. **Thromboprophylaxis** (LMWH) and nasogastric tube if vomiting/drowsy.
11. **Resolution**: ketones below 0.6 mmol/L and pH above 7.3; convert to subcutaneous insulin when eating, overlapping with the infusion by 30–60 minutes.

Definitive

Overview of the step-up approach

1. **Diagnosis (OGTT)** → education + diet + exercise + self-monitoring.
2. Targets not met after **1–2 weeks** → **metformin** (titrate).
3. Targets not met, or metformin not tolerated or contraindicated → **insulin** (add or replace).
4. **Start insulin at once** if fasting glucose is **7.0 or more** at diagnosis, or fasting **6.0–6.9 with macrosomia/polyhydramnios**.

Pre-conception care (pre-existing diabetes)

1. **Plan the pregnancy**: effective contraception until control is good.
2. **Target HbA1c below 48 mmol/mol (6.5%)** if achievable without hypoglycaemia; **strongly advise against pregnancy if HbA1c is above 86 mmol/mol (10%)**.
3. **Folic acid 5 mg orally once daily** from 3 months before conception until 12 weeks.
4. **Stop ACE inhibitors, ARBs and statins** (switch antihypertensives to methyldopa, nifedipine or labetalol).
5. **Switch oral hypoglycaemics** other than metformin to metformin and/or insulin.
6. **Retinal and renal assessment**; treat retinopathy before pregnancy.
7. **Weight loss** if BMI above 27.
8. **Educate** on hypoglycaemia (glucose tablets/sugar, glucagon for type 1) and sick-day rules/ketone testing.

Antenatal care: general measures for all women with hyperglycaemia

1. **Joint diabetes-antenatal clinic** (obstetrician, physician, midwife, dietitian), every 1–2 weeks with the diabetes team.
2. **Education and self-monitoring**: a glucometer and strips where possible.
   - GDM on diet or metformin: fasting and **1-hour post-meal** readings daily.
   - On insulin: fasting, pre-meal, 1-hour post-meal and bedtime readings.
3. **Targets**:

Time	Target (NICE)
Fasting	**below 5.3 mmol/L**
1 hour after meals	**below 7.8 mmol/L**
2 hours after meals	**below 6.4 mmol/L**

Time	Target (NICE)
On insulin: never below	**4.0 mmol/L**
HbA1c (pre-existing)	Below 48 mmol/mol (6.5%) if achievable safely

4. **Aspirin 75–150 mg orally at night from 12 weeks until 36 weeks** for women with **pre-existing** diabetes (a high-risk factor for pre-eclampsia). Uganda commonly uses 75 mg.
5. **Folic acid 5 mg** through the first trimester (pre-existing).
6. **Monitor BP and urine protein** each visit.
7. **Fetal surveillance**: dating scan, anomaly and cardiac scan, growth scans, fetal movement awareness, CTG as indicated.

Diet and exercise (first line for GDM)

- **Regular meals**: 3 meals and 2–3 small snacks, including a bedtime snack if on insulin.
- **Carbohydrate quality and quantity**: replace refined carbohydrates (white bread, sweet tea, sodas, sweet bananas in large amounts, white rice in large portions) with **low-glycaemic-index foods**: whole grains (millet, sorghum, brown rice), **matooke, sweet potatoes, cassava and posho in moderate portions** paired with **beans, groundnuts, greens (dodo, nakati, sukuma), fish, eggs**.
- Spread carbohydrate through the day; avoid large single loads.
- **No sugar in tea**; avoid juices and sodas.
- **Exercise**: 30 minutes of walking daily, ideally **10–15 minutes after each meal**, lowers post-prandial glucose.
- Aim for appropriate, not excessive, weight gain; **do not advise weight-loss diets or fasting** in pregnancy (ketosis harms the fetus).

Metformin

- **Start metformin** if targets are not met after **1–2 weeks** of diet and exercise (dose in the drug table).
- **Add insulin** if targets are still not met on the maximum tolerated dose, or if metformin is contraindicated or not tolerated.
- **Glibenclamide** is a second-line option where insulin is refused, but it crosses the placenta and causes more macrosomia and neonatal hypoglycaemia than insulin or metformin.

Insulin regimens

Insulins used in Uganda: **soluble (regular) human insulin** (e.g. Actrapid), **isophane/NPH intermediate-acting** (e.g. Insulatard), and **premixed 30/70** (30% soluble, 70% isophane; e.g. Mixtard). Analogues (aspart, lispro, detemir, glargine) are used where available.

Regimen	How	Best for
Bedtime isophane (NPH) only	Start about **0.1–0.2 units/kg** subcutaneously at 22:00	GDM with **raised fasting** glucose only
Pre-meal soluble insulin only	Start about **2–4 units** subcutaneously 30 minutes before the meal(s) that cause high post-meal readings	GDM with **raised post-meal** glucose only
Basal-bolus (preferred for type 1 and most type 2)	Isophane at bedtime (± morning) + soluble before each meal	Most flexible; best control
Twice-daily premixed 30/70	Total daily dose split **2/3 before breakfast and 1/3 before supper**, 30 minutes before meals	Practical where multiple injections are not feasible

- **Starting total daily dose (if no previous insulin)**: about **0.3–0.5 units/kg/day** in pregnancy (lower, around 0.2–0.3 units/kg/day, for mild GDM); requirements rise through the third trimester (often up to 0.8–1.0 unit/kg/day by term in type 2 diabetes).
- **Titrate** by about **2 units (or 10%) every 2–3 days** based on the reading the dose controls:

- Fasting high: increase the bedtime isophane.
- Post-breakfast high: increase the morning soluble.
- Pre-supper high: increase the morning isophane (twice-daily regimen) or lunch soluble.

> **DRUG DOSING**
> **Insulin safety**: write "units" in full, never "U" (misread as zero). Use the correct syringe for the vial strength (U-100 with U-100 syringes). Premixed insulin must be rolled, not shaken. Store unopened insulin in the fridge; in-use vials can be kept cool (clay pot) for 4 weeks.

Pre-existing type 2 diabetes

- Continue **metformin**; stop other oral agents (sulphonylureas other than glibenclamide, DPP-4 inhibitors, SGLT2 inhibitors, GLP-1 agonists) and switch to insulin where needed.
- Expect insulin requirements to **rise sharply after 20–24 weeks**.

Pre-existing type 1 diabetes

- Basal-bolus insulin; **hypoglycaemia risk is highest in the first trimester** (nausea, increased sensitivity): reduce doses, carry glucose, give the family **glucagon**.
- **Test for ketones** when unwell or glucose above 11 mmol/L.

Antenatal corticosteroids

- Give if preterm birth is expected (dexamethasone 6 mg IM 12-hourly for 4 doses or betamethasone 12 mg IM 24-hourly for 2 doses, per UCG/WHO criteria).
- **Increase insulin** (or start an IV insulin infusion) and monitor capillary glucose 2–4-hourly for 72 hours.

Fetal surveillance and timing of delivery

Group	Fetal monitoring	Timing of delivery (NICE; widely adopted)
Type 1 or type 2 diabetes	Anomaly/cardiac scan; growth scans every 4 weeks from 28 to 36 weeks	**37+0 to 38+6 weeks** (earlier if complications: poor control, pre-eclampsia, macrosomia, growth restriction, fetal compromise)
GDM on insulin/metformin	Growth scans from diagnosis (e.g. 28, 32, 36 weeks)	Many units deliver at about **38–39 weeks**; ACOG advises **39+0 to 39+6** for well-controlled medication-treated GDM
GDM, diet-controlled, well controlled	Clinical SFH ± scan	**No later than 40+6 weeks** (ACOG: 39+0 to 40+6)

> **UGANDA CONTEXT**
> In many Ugandan referral hospitals, women with **pre-existing diabetes or insulin-treated GDM are admitted around 36–37 weeks** for glucose control and **delivered at 38 weeks**, because home glucose monitoring and fetal surveillance are limited. Follow your unit protocol.

Mode of delivery

- **Vaginal delivery is preferred** (induction at the planned time).
- **Offer caesarean section** if estimated fetal weight is **4,500 g or more** in a diabetic woman (ACOG), for a previous shoulder dystocia with a large baby, or for usual obstetric indications. Ultrasound EFW is inaccurate (about ±15%) in macrosomia; combine it with clinical judgement.

Intrapartum care

1. **Deliver in a hospital** with theatre, neonatal resuscitation and an experienced obstetrician.

2. **Capillary glucose hourly in established labour**, **target 4–7 mmol/L**. Maternal hyperglycaemia in labour causes **fetal hyperinsulinaemia, fetal acidosis and neonatal hypoglycaemia**.
3. **Continuous CTG** (EFM) for pre-existing diabetes and insulin-treated GDM.
4. **GDM on diet or metformin**: stop metformin at onset of labour; check glucose 1–2-hourly; start an insulin infusion if glucose is above 7 mmol/L on two consecutive readings.
5. **Type 1 diabetes**: start **IV dextrose + variable-rate insulin infusion** from the onset of established labour (or on the morning of induction/CS). Type 2 or GDM on insulin: the same if glucose is not in range.
6. **Anticipate shoulder dystocia**: senior obstetrician present in the second stage, McRoberts ready; **avoid mid-cavity instrumental delivery** with suspected macrosomia; empty the bladder.
7. **Anticipate PPH** (large baby, polyhydramnios, prolonged labour): active management of the third stage, oxytocin ready.
8. **Paediatrician/neonatal team informed**.

Example variable-rate IV insulin infusion ("sliding scale")

Set-up: **50 units soluble insulin in 50 mL 0.9% sodium chloride (1 unit/mL)** by syringe driver, **plus** a separate infusion of **5% or 10% dextrose (with 20 mmol KCl per litre if potassium is normal) at about 100 mL/hour** through the same cannula with a one-way valve. Check capillary glucose **hourly**.

Capillary glucose (mmol/L)	Insulin rate (units/hour)
below 4.0	**Stop insulin for 20 minutes (type 1: reduce to 0.5), treat hypoglycaemia, recheck in 15 minutes, call the doctor**
4.0–5.5	0.5–1
5.6–7.0	1–2
7.1–8.5	2–3
8.6–11.0	3–4
above 11.0	4–6; check ketones and call the doctor

> **NOTE**
> This is an **example only**: every hospital has its own scale. Always keep dextrose running with the insulin in type 1 diabetes (to prevent ketosis), and **never stop insulin completely in type 1 diabetes** for long.

When no syringe driver is available: GKI (Alberti) infusion. 500 mL of 10% dextrose + 10 units of soluble insulin + 10 mmol potassium chloride, run at **100 mL/hour** (2 units of insulin/hour). Check glucose hourly; if above 10 mmol/L make a new bag with 15 units; if below 4, make a new bag with 5 units. It is safe because insulin and glucose run together, and it is simple enough for a busy labour ward.

Elective caesarean section or induction

- Put her **first on the morning list**.
- **Night before**: usual evening insulin (consider reducing the bedtime isophane by about a quarter in type 1).
- **Morning**: nil by mouth, **omit the morning short-acting insulin and metformin**, start the IV insulin–dextrose infusion, check glucose hourly.
- Give **antenatal corticosteroids** only if delivering before 37 weeks per guidelines, with extra insulin.
- **Regional anaesthesia** preferred. Antibiotic prophylaxis and thromboprophylaxis as usual.

Drug table

Drug	Dose (adult)	Route	Frequency	Duration	Key cautions
Metformin	Start **500 mg** with the evening meal (or twice daily); increase by 500 mg every few days to 1 week; usual maximum **2,000 mg/day** (up to 2,500 mg/day in some protocols)	Oral	Once to three times daily with meals	Until delivery (GDM); continue postpartum in type 2	Crosses the placenta but appears safe; less maternal weight gain and less neonatal hypoglycaemia than insulin. GI upset: titrate slowly, with food. Avoid if eGFR below 30, hepatic failure, sepsis/shock. **Dolutegravir raises metformin levels** (keep dose low on TLD). Stop at onset of labour; safe in breastfeeding
Isophane (NPH) insulin	Bedtime-only start **0.1–0.2 units/kg** at 22:00	SC	Nightly (± morning)	Titrate ~2 units every 2–3 days	Hypoglycaemia; reduce bedtime dose by about a quarter the night before surgery in type 1
Soluble insulin (pre-meal)	Start **2–4 units** 30 min before the high meal(s)	SC	Before meals	Titrate ~2 units every 2–3 days	Write "units" in full
Premixed 30/70 insulin	Total daily dose (0.3–0.5 units/kg/day; 0.2–0.3 for mild GDM) split **2/3 before breakfast, 1/3 before supper**	SC	Twice daily, 30 min before meals	Throughout pregnancy	Roll, do not shake
Soluble insulin (variable-rate infusion, labour)	50 units in 50 mL 0.9% NaCl (1 unit/mL); rate by sliding scale	IV	Continuous	Labour / peri-operative	With dextrose ± KCl running; hourly glucose
GKI (Alberti)	500 mL 10% dextrose + 10 units soluble insulin + 10 mmol KCl	IV	100 mL/h	Labour / peri-operative	Adjust insulin in the bag (15 units if above 10; 5 units if below 4)
Soluble insulin (DKA)	**0.1 unit/kg/h** fixed rate	IV	Continuous	Until ketones below 0.6 and pH above 7.3	Continue usual long-acting insulin
0.9% sodium chloride (DKA)	500 mL over 10–15 min if SBP below 90; then e.g. 1 L/1 h, 1 L/2 h, 1 L/2 h, 1 L/4 h, 1 L/4 h	IV	As listed	Reassess	Caution in pre-eclampsia, cardiac or renal disease
Potassium chloride (DKA)	40 mmol per litre if K^+ 3.5–5.5; none if above 5.5; senior review if below 3.5	IV	In each litre from the second	Guided by K^+	

Drug	Dose (adult)	Route	Frequency	Duration	Key cautions
10% dextrose (DKA)	125 mL/h	IV	When glucose below 14 mmol/L	Until ketones clear	Run alongside saline
Glucose / sugar (hypoglycaemia)	15–20 g (3–4 teaspoons in water or 150–200 mL soda)	Oral	Once; recheck in 15 min	Then a snack	
20% dextrose (hypoglycaemia)	75–100 mL (or 10% dextrose 150–200 mL; 50% 25–50 mL only if nothing else, large vein)	IV	Once; recheck		50% is irritant
Glucagon	1 mg	IM	Once		Give to the family of women with type 1
Folic acid (pre-existing)	**5 mg**	Oral	Daily	3 months before conception to 12 weeks	
Aspirin (pre-existing)	**75–150 mg** (Uganda commonly 75 mg)	Oral	At night	**12 to 36 weeks**	
Dexamethasone	6 mg	IM	12-hourly	4 doses	Increase insulin; glucose 2–4-hourly for 72 h
Betamethasone	12 mg	IM	24-hourly	2 doses	As above
Glibenclamide	Per protocol	Oral			Second line only; crosses placenta; more macrosomia and neonatal hypoglycaemia
10% dextrose (neonate)	**2 mL/kg** bolus, then maintenance infusion	IV	Once, then infusion		Symptomatic neonatal hypoglycaemia
Nifedipine (tocolysis if needed)	See preterm labour topic	Oral			Preferred over beta-agonists in diabetes

Methyldopa, nifedipine or labetalol replace ACE inhibitors and ARBs for hypertension (doses in the hypertension topic).

Referral criteria

Level	What should happen	Refer up when
HC III	Identify risk factors, urine dipstick, capillary glucose; health education; treat hypoglycaemia	High-risk women and those with glycosuria → HC IV for OGTT
HC IV	OGTT (if strips/glucose available), start diet and metformin under a doctor, monitor; manage emergencies (hypoglycaemia)	**Insulin-requiring or pre-existing diabetes** → hospital for joint care; **DKA** → refer after starting fluids
General Hospital	Joint diabetes-antenatal clinic, insulin initiation and titration, growth scans, CTG, induction/CS, IV insulin in labour, neonatal care	DKA needing HDU, pre-existing diabetes with complications, fetal anomaly → RRH
Regional Referral Hospital	As above plus DKA management, HbA1c, retinal assessment, detailed anomaly scan, neonatal unit	Fetal cardiac anomaly needing paediatric cardiology/surgery, ICU → NRH
National Referral Hospital	Paediatric cardiology/surgery, ICU, blood gas	n/a

The source notes grouped general, regional and national referral hospitals together; the split is adapted Verify *against UCG 2023 levels of care*.

Monitoring and follow-up

- **Antenatal**: glucose profile reviewed every 1–2 weeks; BP and urine each visit; HbA1c each trimester (pre-existing); growth scans by group; fetal movements; insulin requirement (a fall of more than 15–20% in the third trimester = assess the fetus).
- **Labour**: capillary glucose hourly (4–7 mmol/L); continuous CTG if pre-existing or insulin-treated.
- **DKA**: glucose and ketones hourly; K^+ and bicarbonate at 1, 2 and 4 hours then 4-hourly; fluid balance.

Postpartum care

The mother

Group	Action
GDM	**Stop all glucose-lowering treatment immediately after delivery.** Check capillary glucose (e.g. fasting next day) before discharge to exclude persisting hyperglycaemia
Type 1 diabetes	**Reduce insulin immediately** to the pre-pregnancy dose (or about 50% of the late-pregnancy dose); stop the infusion once eating, with an overlap with subcutaneous insulin. Risk of **hypoglycaemia**, especially when breastfeeding: eat before or during feeds
Type 2 diabetes	Return to the pre-pregnancy regimen; **metformin (and glibenclamide) are safe in breastfeeding**. Avoid other oral agents while breastfeeding
All	Encourage **breastfeeding** (lowers the mother's future diabetes risk and the baby's obesity risk), **contraception** and pre-conception planning, thromboprophylaxis as per risk

Follow-up after GDM

1. **Test at 6–13 weeks postpartum**: **fasting plasma glucose** (NICE) or a **75 g OGTT** (WHO/ACOG, at 4–12 weeks). After 13 weeks, HbA1c can be used.
2. Interpretation (NICE): fasting below 6.0 = low risk; **6.0–6.9 = high risk** (lifestyle, prevention programme); **7.0 or more = likely type 2 diabetes** (confirm).
3. **Annual HbA1c or fasting glucose for life**.

4. Lifestyle: weight control, diet, activity.
5. **Early OGTT in the next pregnancy** (recurrence is about 30–50%).

The baby

1. **Skin-to-skin and breastfeed within 30–60 minutes** of birth, then **every 2–3 hours**.
2. **Check blood glucose at 2–4 hours after birth, before the second feed**, then before feeds for at least 24 hours.
3. **Treat hypoglycaemia**: many Ugandan neonatal protocols treat glucose **below 2.6 mmol/L** (NICE uses below 2.0 mmol/L on two consecutive readings despite feeding support, or any low reading with abnormal clinical signs). Symptomatic babies (jittery, lethargic, convulsions, poor feeding) need **IV 10% dextrose 2 mL/kg bolus**, then a maintenance infusion, and admission.
4. Look for **respiratory distress, jaundice, polycythaemia, hypocalcaemia, birth trauma (Erb's palsy, fractured clavicle) and congenital anomalies**.
5. **Do not transfer to the neonatal unit for monitoring alone** if well: keep mother and baby together.

10. Complications

Disease-related

Maternal	Fetal / neonatal
Pre-eclampsia	**Congenital anomalies** (pre-existing: cardiac, NTD, caudal regression)
Polyhydramnios leading to preterm labour, malpresentation, cord prolapse, abruption, PPH	**Miscarriage** (pre-existing)
Infections: UTI, pyelonephritis, candidiasis, endometritis, wound infection	**Macrosomia** leading to **shoulder dystocia**, brachial plexus injury (Erb's palsy), fractured clavicle, asphyxia
DKA	**Stillbirth** (late, unexplained)
Worsening **retinopathy** and **nephropathy**	**Neonatal hypoglycaemia**, hypocalcaemia, hypomagnesaemia
Operative delivery, perineal trauma, PPH	**Respiratory distress syndrome** (delayed surfactant maturation), transient tachypnoea
Future **type 2 diabetes** (up to about 50% within 5–10 years), metabolic syndrome, cardiovascular disease	**Polycythaemia**, **jaundice**, hypertrophic cardiomyopathy
Recurrent GDM in future pregnancies	Growth restriction (pre-existing diabetes with vasculopathy)
	Long term: childhood obesity and type 2 diabetes

Treatment-related

Complication	Cause	Action
Maternal hypoglycaemia	Insulin (especially type 1, first trimester, breastfeeding); GDM treatment continued after delivery	15–20 g fast sugar or IV dextrose/glucagon; stop GDM treatment at delivery
Retinopathy progression	Rapid tightening of control	Retinal assessment at booking and 28 weeks
Hyperglycaemia, DKA	Steroids, beta-agonist tocolysis	Increase insulin; use nifedipine
GI upset; raised levels with dolutegravir	Metformin	Titrate slowly; lower dose on TLD

Complication	Cause	Action
Macrosomia, neonatal hypoglycaemia	Glibenclamide	Second line only
Hypokalaemia, pulmonary oedema	DKA treatment	K^+-guided replacement; cautious fluids

11. Prognosis

- With tight control, the risks of macrosomia, shoulder dystocia, stillbirth and neonatal hypoglycaemia fall.
- **GDM** usually resolves after delivery, but **up to about 50% develop type 2 diabetes within 5–10 years**; **recurrence in a later pregnancy is about 30–50%**.
- **Pre-existing diabetes**: anomaly and miscarriage risk rise with periconception HbA1c; very high above 86 mmol/mol (10%).
- **DKA**: fetal loss **10–35%**.
- Children: long-term risk of obesity and type 2 diabetes.

12. Prevention

- **Primary**: healthy weight before pregnancy, diet and activity; universal (or at least risk-based) OGTT at 24–28 weeks.
- **Pre-existing diabetes**: pre-conception care (HbA1c target, folic acid 5 mg, stop ACEi/ARBs/statins, retinal and renal assessment, contraception until control is good).
- **Of complications**: tight control, growth scans, timed delivery before 40 weeks, senior at delivery, neonatal glucose checks.
- **Of DKA**: sick-day rules, ketone testing, avoid beta-agonists, extra insulin with steroids, treat infections early.
- **After GDM**: breastfeeding, lifestyle, test at 6–13 weeks and **annually for life**, early OGTT in the next pregnancy.

13. Special Populations

- **Type 1 diabetes**: first-trimester hypoglycaemia (reduce doses, glucagon for the family), ketone testing, IV insulin–dextrose from established labour, never stop insulin for long, reduce insulin immediately postpartum.
- **Type 2 diabetes**: continue metformin, switch other agents, insulin requirements rise after 20–24 weeks.
- **HIV on TLD**: dolutegravir raises metformin levels; keep the metformin dose low.
- **Anaemia, sickle cell disease, recent transfusion**: HbA1c falsely low; rely on glucose profiles.
- **Pre-eclampsia or renal/cardiac disease in DKA**: cautious fluids (pulmonary oedema).
- **Nephropathy**: metformin contraindicated if eGFR below 30; predicts pre-eclampsia and FGR.
- **Preterm labour**: give steroids with extra insulin; use nifedipine, not beta-agonists.
- **Breastfeeding mothers**: metformin and glibenclamide are safe; type 1 women eat before or during feeds.

14. Key Points

High-yield points

- **Glucose crosses the placenta, insulin does not** (Pedersen).
- **75 g OGTT at 24–28 weeks**; WHO 2013 **5.1 / 10.0 / 8.5**, any one value.

- **HbA1c and glycosuria do not diagnose GDM.**
- Targets: fasting below 5.3, 1 h below 7.8, 2 h below 6.4; never below 4.0.
- Diet → metformin → insulin; insulin at once if fasting 7.0 or more.
- High fasting → bedtime isophane; high post-meal → diet or pre-meal soluble.
- Pre-existing: folic acid 5 mg, aspirin from 12 weeks, anomaly scan.
- Deliver type 1/2 at 37+0–38+6; diet GDM by 40+6; glucose 4–7 in labour.
- **Stop GDM treatment at delivery**; neonatal glucose at 2–4 h; test the mother at 6–13 weeks, then yearly.
- DKA: fluids, insulin 0.1 unit/kg/h, K^+, dextrose; **treat the mother, not the CTG**.

Numbers & doses

Item	Value
OGTT timing	**24–28 weeks**; early (and repeat at 24–28) if previous GDM / very high risk
OGTT prep	**8–14 h fast**, **75 g anhydrous glucose in 250–300 mL water** over 5 min
WHO 2013 GDM (any one)	Fasting **5.1–6.9**, 1 h **10.0 or more**, 2 h **8.5–11.0** mmol/L
Overt diabetes in pregnancy	Fasting **7.0 or more**, 2 h **11.1 or more**, random **11.1 or more** with symptoms; HbA1c **48 mmol/mol (6.5%) or more** at booking
NICE GDM diagnosis	Fasting **5.6 or more** or 2 h **7.8 or more**
Targets (NICE)	Fasting **below 5.3**; 1 h **below 7.8**; 2 h **below 6.4**; never below **4.0**
Glycosuria trigger for OGTT	**2+ once or 1+ twice**
Metformin	Start **500 mg** with a meal, titrate to usual max **2,000 mg/day**
Starting insulin (no previous insulin)	Total **0.3–0.5 units/kg/day** (0.2–0.3 for mild GDM); bedtime isophane **0.1–0.2 units/kg**; titrate **~2 units every 2–3 days**
Premixed 30/70 split	**2/3 before breakfast, 1/3 before supper**
Insulin at diagnosis	Fasting **7.0 or more** (or 6.0–6.9 with macrosomia/polyhydramnios)
Hypoglycaemia	Below **4.0**: **15–20 g** fast sugar; unconscious: **20% dextrose 75–100 mL IV** or **glucagon 1 mg IM**
Pre-existing diabetes	**Folic acid 5 mg** daily; **aspirin 75–150 mg** from **12 to 36 weeks**; HbA1c target **below 48 mmol/mol**; avoid pregnancy if **above 86 mmol/mol (10%)**
Delivery timing	Type 1/2: **37+0 to 38+6**; diet-controlled GDM: **no later than 40+6**
Offer CS	EFW **4,500 g or more** (ACOG)
Intrapartum glucose	**4–7 mmol/L**, **hourly**
GKI (Alberti)	**500 mL 10% dextrose + 10 units soluble insulin + 10 mmol KCl at 100 mL/h**
DKA diagnosis	Ketones **3.0 or more** (urine **2+ or more**) + bicarbonate **below 15** or pH **below 7.3**
DKA insulin / dextrose	**0.1 unit/kg/h** fixed rate; add **10% dextrose 125 mL/h** when glucose **below 14**
Neonate	Feed within **30–60 min**; glucose at **2–4 h** (before 2nd feed); treat below **2.6** (many Ugandan units) / **2.0** (NICE); symptomatic: **10% dextrose 2 mL/kg IV**
Postpartum GDM test	**6–13 weeks** (fasting glucose or OGTT), then **yearly**; recurrence **30–50%**

Exam traps

- **"HbA1c of 5.4% rules out GDM."** Wrong: HbA1c **does not diagnose GDM** (it falls in pregnancy and is unreliable with anaemia/sickle cell). Do an **OGTT**.

- **"A random glucose of 6.8 excludes GDM."** Wrong: a random glucose cannot rule it out; do an OGTT at 24–28 weeks.
- **"Glycosuria 2+ confirms diabetes."** Wrong: the renal threshold falls in pregnancy; glycosuria is a **trigger for an OGTT**.
- **"Continue metformin and insulin after delivery in GDM."** Wrong: **stop all GDM treatment at delivery** (the placenta has gone) and test at 6–13 weeks.
- **"Abnormal CTG in DKA: category 1 caesarean."** Wrong: **correct the mother's acidosis first**; the CTG usually recovers.
- **"Salbutamol for tocolysis."** Wrong in diabetes: it causes hyperglycaemia and DKA; use **nifedipine**.
- **"Withhold steroids because she is diabetic."** Wrong: give them if indicated and **increase insulin** with 2–4-hourly glucose checks for 72 hours.
- **"GDM causes congenital anomalies."** Wrong: GDM starts after organogenesis. **Pre-existing** (or overt, undiagnosed type 2) diabetes does.
- **"Borderline GDM."** There is no borderline in WHO 2013: one abnormal value makes the diagnosis.

Questions seniors ask

On the ward round

1. **"Read me this OGTT: fasting 5.3, 1 h 9.4, 2 h 7.4."** GDM by WHO 2013 (fasting 5.1 or more); not GDM by NICE. Uganda uses WHO 2013.
2. **"Her fasting readings are 6.0 on metformin 2 g/day but post-meal readings are fine. What do you add?"** **Bedtime isophane** at 22:00, about 0.1–0.2 units/kg, titrated by 2 units every 2–3 days to fasting below 5.3.
3. **"What does a falling insulin requirement in the third trimester mean?"** Possible **placental insufficiency** (more than about 15–20% fall): assess fetal wellbeing.
4. **"How will you manage her glucose in labour?"** Hourly capillary glucose, target 4–7 mmol/L; stop metformin; IV insulin–dextrose infusion (or GKI) if above 7 on two readings, and from established labour in type 1.
5. **"What do you tell the midwife about the baby?"** Skin-to-skin, breastfeed within 30–60 minutes and every 2–3 hours, glucose at 2–4 hours before the second feed, watch for jitteriness, respiratory distress, jaundice and birth injury.
6. **"Why is DKA different in pregnancy?"** Reduced bicarbonate buffer and accelerated starvation: it occurs faster and at lower glucose (even below 11 mmol/L); fetal loss is 10–35%.
7. **"When will you deliver a woman with type 1 diabetes, and how?"** 37+0 to 38+6 weeks, vaginal delivery by induction unless there is an obstetric indication or EFW 4,500 g or more; senior present for shoulder dystocia.
8. **"What follow-up does she need after GDM?"** Fasting glucose or OGTT at 6–13 weeks, then yearly HbA1c/fasting glucose for life, lifestyle advice, and an early OGTT in the next pregnancy.

Likely viva questions

1. **What hormones cause insulin resistance in pregnancy?** Human placental lactogen (the main one), placental growth hormone, progesterone, cortisol, prolactin (and TNF-α from the placenta).
2. **State the WHO 2013 OGTT criteria.** Fasting 5.1–6.9, 1-hour 10.0 or more, 2-hour 8.5–11.0 mmol/L; any one value.
3. **Explain the Pedersen hypothesis.** Maternal glucose crosses the placenta; fetal hyperglycaemia stimulates fetal hyperinsulinaemia, which causes macrosomia, organomegaly and, after cord clamping, neonatal hypoglycaemia.
4. **Why is GDM not associated with congenital anomalies?** It arises after organogenesis (after the first trimester).

5. **Which women in Uganda should be screened?** Ideally all women at 24–28 weeks (universal screening); at minimum those with BMI above 30, previous GDM, previous baby 4 kg or more, first-degree relative with diabetes, previous unexplained stillbirth, polyhydramnios, glycosuria, PCOS, age 35 or more. Previous GDM: test early and again at 24–28 weeks.
6. **What are the glucose targets?** Fasting below 5.3, 1-hour below 7.8, 2-hour below 6.4 mmol/L.
7. **When do you start insulin at diagnosis?** Fasting glucose 7.0 or more (or 6.0–6.9 with macrosomia/polyhydramnios).
8. **When should a woman with type 1 diabetes deliver?** 37+0 to 38+6 weeks.
9. **What is the intrapartum glucose target?** 4–7 mmol/L, hourly checks.
10. **What happens to GDM treatment after delivery?** Stop it; test at 6–13 weeks; annual screening.
11. **Why is DKA dangerous in pregnancy and how is it different?** Occurs faster and at lower glucose (reduced bicarbonate buffering, accelerated starvation); fetal loss up to a third; treat the mother, not the CTG.
12. **Why avoid salbutamol tocolysis in diabetes?** Beta-agonists raise glucose and can trigger DKA.

Mnemonics

Placental anti-insulin hormones: "Placental Hormones Can Produce Problems"

- **P**lacental lactogen (hPL)
- **H**ormone, placental **G**rowth
- **C**ortisol
- **P**rogesterone
- **P**rolactin

WHO 2013 OGTT: "5.1, 10, 8.5" (think "**5-10-8**": fasting, 1 hour, 2 hours).

NICE targets: "5.3, 7.8, 6.4" (fasting, 1 h, 2 h). NICE diagnosis: "**5.6 / 7.8**".

Fetal complications of hyperinsulinaemia: "MACROSOMIC"

- **M**acrosomia and shoulder dystocia
- **A**nomalies (pre-existing only)
- **C**ardiomyopathy (septal hypertrophy)
- **R**espiratory distress syndrome
- **O**rganomegaly
- **S**tillbirth
- **O**ver-distended uterus (**polyhydramnios**, from fetal polyuria)
- **M**agnesium and calcium low
- **I**cterus (jaundice) and polycythaemia
- **C**old, jittery, hypoglycaemic newborn

DKA management: "FIKD" (in order): **F**luids, **I**nsulin 0.1 unit/kg/h, **K**$^{+}$ (potassium), **D**extrose when glucose below 14.

OSCE tips

- **Counselling a woman newly diagnosed with GDM**: explain what it is (placental hormones, not her fault), why it matters (big baby, stillbirth, baby's low sugar), what she will do (diet, walking, checking sugars, maybe tablets or injections), delivery plan, and **lifelong risk of diabetes** with the 6–13-week test. Check understanding.
- **Insulin prescribing station**: write drug name in full, dose in **units** (never "U"), route, timing relative to meals, and hypoglycaemia instructions.

- **OGTT station**: explain the preparation (3 days normal diet, 8–14 h fast), 75 g glucose in 250–300 mL over 5 minutes, sitting quietly, samples at 0, 1 and 2 hours, and the WHO 2013 cut-offs.

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SELF-TEST

Flashcards

27 cards. Cover the right-hand column and test yourself.

#	Question	Answer
1	Which placental hormones cause insulin resistance in pregnancy?	**Human placental lactogen** (main), placental growth hormone, **progesterone**, **cortisol**, **prolactin** (and placental TNF-alpha). Resistance rises from the second trimester and disappears within hours of delivering the placenta.
2	Why do women with type 1 diabetes get more hypoglycaemia in the first trimester?	Early pregnancy **increases insulin sensitivity** (and fasting glucose falls), and nausea/vomiting reduces intake. Insulin doses often need reducing.
3	State the WHO 2013 75 g OGTT thresholds for GDM.	Any one of: **fasting 5.1–6.9 mmol/L**, **1-hour 10.0 mmol/L or more**, **2-hour 8.5–11.0 mmol/L**.
4	What OGTT or fasting values indicate overt diabetes in pregnancy (WHO 2013)?	**Fasting 7.0 mmol/L or more**, **2-hour 11.1 or more**, or **random 11.1 or more with symptoms**.
5	NICE diagnostic thresholds for GDM?	**Fasting 5.6 mmol/L or more** or **2-hour 7.8 mmol/L or more** on a 75 g OGTT.
6	Explain the Pedersen hypothesis in one line.	**Glucose crosses the placenta, insulin does not**: fetal hyperglycaemia causes **fetal hyperinsulinaemia**, which drives macrosomia, organomegaly, polyhydramnios, delayed lung maturity and neonatal hypoglycaemia.
7	Why is macrosomia in diabetes associated with more shoulder dystocia?	Insulin-driven growth is **asymmetric**: fat is deposited on the **shoulders, chest and abdomen**, which outgrow the head, so the shoulders impact even at lower birth weights.
8	Why does a baby of a diabetic mother become hypoglycaemic after birth?	Its pancreas has been making **excess insulin** in response to maternal glucose. At cord clamping the glucose supply stops but hyperinsulinaemia persists for hours to days.
9	Glucose targets in pregnancy (NICE)?	**Fasting below 5.3**, **1-hour post-meal below 7.8**, or **2-hour post-meal below 6.4 mmol/L**; on insulin keep above **4.0 mmol/L**.
10	Which women should have an OGTT in Uganda (risk-based)?	BMI above 30, **previous GDM** (test early and again at 24–28 weeks), **previous baby 4 kg or more**, first-degree relative with diabetes, previous unexplained stillbirth, polyhydramnios, **glycosuria 2+ once or 1+ twice**, PCOS, age 35 or more. Universal screening at 24–28 weeks is ideal.
11	First-line treatment of GDM and when to escalate?	**Diet and exercise** with self-monitoring. If targets are not met in **1–2 weeks**, add **metformin**; if still not met, add **insulin**. Start **insulin immediately** if fasting glucose is **7.0 or more** at diagnosis.
12	Metformin dosing in pregnancy?	Start **500 mg orally** with the evening meal (or twice daily), increase by 500 mg every few days to a week, usual maximum **2,000 mg/day** in divided doses with meals. Keep low in women on dolutegravir.
13	Folic acid dose for a woman with pre-existing diabetes planning pregnancy?	**5 mg orally once daily** from 3 months before conception until 12 weeks' gestation.
14	Which drugs must a woman with pre-existing diabetes stop before/at conception?	**ACE inhibitors, ARBs and statins**; oral hypoglycaemics other than metformin (switch to insulin/metformin).
15	Timing of delivery for type 1/type 2 diabetes and for diet-controlled GDM (NICE)?	Type 1/2: **37+0 to 38+6 weeks**. Diet-controlled GDM: **no later than 40+6 weeks**. Earlier if complications.

#	Question	Answer
16	Intrapartum capillary glucose target and frequency?	**4–7 mmol/L**, checked **hourly** in established labour. Use an IV insulin–dextrose infusion if above range (always for type 1).
17	Describe a GKI (Alberti) infusion.	**500 mL of 10% dextrose + 10 units soluble insulin + 10 mmol KCl at 100 mL/hour**; adjust insulin in the next bag (5 or 15 units) according to hourly glucose. Useful where no syringe driver.
18	What do you do with GDM treatment after delivery?	**Stop all glucose-lowering treatment** immediately, check glucose before discharge, then **fasting glucose at 6–13 weeks** (or OGTT at 4–12 weeks), and **annual screening for life**.
19	Why does DKA occur at lower glucose levels in pregnancy?	Compensated **respiratory alkalosis** lowers serum **bicarbonate** (less buffer), plus **accelerated starvation** and insulin resistance, so ketoacidosis develops faster, sometimes with glucose below 11 mmol/L.
20	Core DKA treatment in pregnancy?	0.9% saline, **fixed-rate IV insulin 0.1 unit/kg/h**, **potassium 40 mmol/L** if K^+ 3.5–5.5, **10% dextrose 125 mL/h when glucose falls below 14**, treat the trigger, left lateral tilt. **Do not rush to CS**; CTG usually recovers as the mother is corrected.
21	Why should salbutamol be avoided for tocolysis in a diabetic woman?	Beta-agonists cause **hyperglycaemia** and can precipitate **DKA**. Nifedipine is preferred.
22	Neonatal glucose monitoring after a diabetic pregnancy?	Feed within **30–60 minutes**, then every 2–3 hours; check glucose **2–4 hours after birth (before the second feed)** and before feeds for 24 hours. Treat low glucose per protocol (many Ugandan units treat below 2.6 mmol/L).
23	OGTT result: fasting 5.3, 1 h 9.4, 2 h 7.4 mmol/L. Diagnosis by WHO 2013 and by NICE?	**WHO 2013: GDM** (fasting 5.1 or more; any one value is enough). **NICE: not GDM** (needs fasting 5.6 or more or 2 h 7.8 or more). Uganda uses **WHO 2013**.
24	On metformin 2 g/day, fasting readings are 5.9–6.1 but all 1-hour post-meal readings are below 7.8. What do you change?	Add **bedtime isophane (NPH) insulin at 22:00**, starting about **0.1–0.2 units/kg** (lower end first), and increase by **~2 units every 2–3 days** until fasting is **below 5.3 mmol/L** without readings below 4.0.
25	Which urine dipstick results in a diabetic pregnant woman need action, and what action?	**Glucose** 2+ once or 1+ twice → OGTT (if not yet diagnosed). **Ketones** 2+ or more when unwell → check glucose, bicarbonate/blood gas: think **DKA**. **Protein** → pre-eclampsia workup. **Nitrites/leucocytes** → urine culture (UTI is a DKA trigger).
26	What are the first actions for a woman with diabetes whose capillary glucose is 2.9 mmol/L and who cannot swallow?	**IV 75–100 mL of 20% dextrose** (or 150–200 mL of 10%), or **glucagon 1 mg IM** if no IV access; recheck glucose in 15 minutes, give a snack once awake, and find the cause (insulin dose, missed meal, early-pregnancy sensitivity).
27	Why can HbA1c mislead in Ugandan pregnant women, and what is it used for?	It **falls in pregnancy** (increased red-cell turnover) and is falsely low with **anaemia, recent transfusion, malaria and sickle cell disease**. It is **not used to diagnose GDM**; use it at booking to detect overt diabetes (**48 mmol/mol / 6.5% or more**) and to monitor known diabetes.

SELF-TEST

Single best answer questions

15 questions. The answers and explanations follow at the end.

1. A 29-year-old G3P2 at 26 weeks, BMI 33, has a 75 g OGTT: fasting 5.3, 1-hour 9.2, 2-hour 7.9 mmol/L. How should this result be interpreted using WHO 2013 criteria?

A Normal; no GDM
B GDM, because the fasting value is 5.1 or more
C GDM only if two values are abnormal
D Overt diabetes in pregnancy
E Impaired glucose tolerance, repeat at 32 weeks

2. A 31-year-old at 28 weeks is diagnosed with GDM on OGTT (fasting 5.4, 2-hour 8.9 mmol/L). She has no complications and the scan shows normal growth. What is the MOST appropriate initial management?

A Start insulin 0.5 units/kg/day
B Start glibenclamide 5 mg daily
C No treatment; repeat OGTT at 34 weeks
D Admit for induction of labour at 34 weeks
E Dietary advice, daily walking and capillary glucose self-monitoring, reviewed in 1–2 weeks

3. A 27-year-old at 30 weeks with GDM has followed diet and exercise for 2 weeks. Her readings: fasting 5.8–6.2, 1-hour post-meal 8.0–8.6 mmol/L. She is on TLD for HIV. What is the BEST next step? Advanced

A Continue diet only and recheck in 4 weeks
B Start premixed insulin 30/70 at 1 unit/kg/day
C Start metformin at a low dose (500 mg with the evening meal) and titrate carefully, noting the interaction with dolutegravir
D Stop dolutegravir and start efavirenz so that metformin can be used
E Induce labour at 32 weeks

4. A 26-year-old at 32 weeks with GDM on metformin 1 g twice daily has persistently high fasting glucose (6.4–6.8 mmol/L) but normal post-meal readings. What is the MOST appropriate change?

A Add isophane (NPH) insulin at bedtime, starting about 0.1–0.2 units/kg and titrating
B Add soluble insulin before each meal
C Increase metformin to 4 g/day
D Add a bedtime snack of sweet porridge
E Stop metformin and start glibenclamide

5. A 24-year-old with type 1 diabetes at 8 weeks has had three episodes of confusion and sweating early in the morning this week. Her insulin doses are unchanged from before pregnancy. What is the MOST likely explanation?

A Diabetic ketoacidosis
B Insulin resistance from human placental lactogen
C Hyperemesis-related hyponatraemia
D Increased insulin sensitivity of early pregnancy with vomiting, causing hypoglycaemia
E Hypothyroidism

6. A 30-year-old with type 2 diabetes on metformin and glimepiride, plus lisinopril for hypertension, attends pre-conception clinic. HbA1c is 9.8% (84 mmol/mol). What is the MOST appropriate advice?

A Conceive now; control will improve in pregnancy

B Use effective contraception while improving control (target HbA1c below 6.5%); stop glimepiride and lisinopril, switching to insulin/metformin and nifedipine or methyldopa; folic acid 5 mg daily from 3 months pre-conception

C Continue current drugs and start folic acid 0.4 mg

D Stop all diabetes medication before conception

E Start aspirin now and conceive

7. A 28-year-old P1 at 38 weeks with insulin-treated GDM is in established labour. Capillary glucose is 9.4 mmol/L. What is the MOST appropriate action?

A Give her usual subcutaneous premixed insulin dose

B Give 50 mL of 50% dextrose

C Recheck glucose after delivery

D Perform an emergency caesarean section

E Start an IV insulin–dextrose (variable-rate or GKI) infusion and check glucose hourly aiming for 4–7 mmol/L

8. A 32-year-old with GDM controlled on metformin 1 g twice daily delivers vaginally. What should happen to her metformin?

A Continue for 6 weeks postpartum

B Halve the dose while breastfeeding

C Stop immediately after delivery and check glucose before discharge

D Switch to insulin until the 6-week check

E Continue for life

9. A 23-year-old with type 1 diabetes at 30 weeks presents with 2 days of vomiting, abdominal pain and rapid deep breathing. Capillary glucose 13.2 mmol/L, urine ketones 3+, bicarbonate 10 mmol/L, K^+ 4.6 mmol/L. CTG shows reduced variability and late decelerations. What is the MOST appropriate immediate management?

Advanced

A IV 0.9% saline, fixed-rate IV insulin 0.1 unit/kg/hour, potassium replacement in fluids, left lateral tilt and fetal monitoring as the mother is stabilised

B Emergency caesarean section for fetal distress

C Subcutaneous soluble insulin 10 units and discharge

D IV sodium bicarbonate 8.4% 100 mL

E Oral rehydration and antiemetics

10. A 35-year-old at 36 weeks with pre-existing type 2 diabetes on insulin notices that her insulin requirement has fallen by 25% over the last 2 weeks without dietary change. What is the MOST important next step?

Advanced

A Reduce insulin and reassure

B Increase carbohydrate intake

C Stop insulin because she is cured

D Assess fetal wellbeing urgently (CTG, growth scan, liquor, umbilical artery Doppler)

E Arrange an OGTT

11. A 26-year-old at 29 weeks with GDM on insulin is admitted in threatened preterm labour. Which management is MOST appropriate? Advanced

A Salbutamol infusion for tocolysis and dexamethasone

B Nifedipine for tocolysis if indicated, dexamethasone 6 mg IM 12-hourly for 4 doses, with increased insulin and 2–4-hourly glucose monitoring

C Withhold steroids because of diabetes

D Dexamethasone and stop insulin

E Magnesium sulphate for tocolysis and withhold steroids

12. A term baby weighing 4.4 kg is born to a mother with poorly controlled GDM. At 3 hours the baby is jittery and feeding poorly; glucose is 1.8 mmol/L. What is the MOST appropriate management?

A Give oral water and recheck in 6 hours

B Start formula feeds only and discharge

C Give 50% dextrose 5 mL/kg IV

D Observe; jitteriness is normal in large babies

E IV 10% dextrose 2 mL/kg bolus then a maintenance dextrose infusion, admit to the neonatal unit and monitor glucose

13. A 30-year-old at 38 weeks with GDM on metformin, well controlled, has an estimated fetal weight of 4.7 kg on ultrasound. What is the BEST plan for delivery? Advanced

A Induction of labour now

B Await spontaneous labour up to 41 weeks

C Offer elective caesarean section

D Deliver by vacuum extraction at full dilatation

E External cephalic version

14. A 33-year-old who had GDM delivered 8 weeks ago. Her fasting plasma glucose today is 6.3 mmol/L. How is this BEST interpreted and managed (NICE)?

A High risk of type 2 diabetes: lifestyle advice, diabetes-prevention support and annual testing

B Normal; no follow-up needed

C Type 2 diabetes: start metformin immediately

D GDM persisting: restart insulin

E Repeat testing only in her next pregnancy

15. At a HC III, a 25-year-old at 26 weeks has glycosuria 2+ on dipstick. Her last baby weighed 4.2 kg. The HC III has no glucose strips or glucose powder. What is the MOST appropriate action?

A Reassure her: glycosuria is normal in pregnancy

B Start metformin empirically

C Start insulin empirically

D Refer to the nearest HC IV/hospital for a 75 g OGTT at 24–28 weeks and explain why

E Advise her to stop eating carbohydrates

Answers & explanations

> **1. B** · GDM, because the fasting value is 5.1 or more
>
> **B is correct.** WHO 2013 requires **any one** value: fasting 5.1–6.9, 1-hour 10.0 or more, or 2-hour 8.5–11.0. Her fasting 5.3 qualifies. **A** ignores the fasting value. **C** describes older two-value criteria (Carpenter-Coustan), not WHO 2013. **D** requires fasting 7.0 or more or 2-hour 11.1 or more. **E**: 'impaired glucose tolerance' is not a pregnancy category, and delaying diagnosis delays treatment.

2. E · Dietary advice, daily walking and capillary glucose self-monitoring, reviewed in 1–2 weeks

E is correct. Mild GDM (fasting below 7.0, no complications) starts with **diet, exercise and self-monitoring**, escalating to metformin if targets are not met in 1–2 weeks. **A**: insulin first-line is for fasting 7.0 or more or complications. **B**: glibenclamide is second-line (crosses the placenta, more macrosomia/neonatal hypoglycaemia). **D**: no indication for preterm delivery. **C**: untreated GDM increases macrosomia and stillbirth.

3. C · Start metformin at a low dose (500 mg with the evening meal) and titrate carefully, noting the interaction with dolutegravir

C is correct. Targets are not met after 1–2 weeks, so add **metformin**; **DTG raises metformin levels**, so start low and titrate, monitoring for side effects. **A** delays needed treatment. **B**: insulin is reasonable if metformin fails or is contraindicated, but 1 unit/kg/day is an excessive starting dose that risks hypoglycaemia. **D**: changing effective ART is unnecessary and risky. **E**: no indication for preterm delivery.

4. A · Add isophane (NPH) insulin at bedtime, starting about 0.1–0.2 units/kg and titrating

A is correct. Fasting hyperglycaemia reflects overnight hepatic glucose output, controlled by **basal (bedtime isophane) insulin**. **B**: pre-meal soluble insulin treats post-meal peaks, which are already controlled. **C**: 4 g exceeds the usual maximum (2–2.5 g). **D**: a sugary snack would worsen fasting glucose. **E**: glibenclamide is inferior and does not specifically target the fasting problem.

5. D · Increased insulin sensitivity of early pregnancy with vomiting, causing hypoglycaemia

D is correct. The first trimester brings **increased insulin sensitivity** and nausea/reduced intake, so **hypoglycaemia** is common in type 1 diabetes. Reduce doses, carry glucose, supply glucagon. **A**: DKA causes vomiting, abdominal pain and drowsiness, not recurrent early-morning sweating episodes that resolve. **C**: hyponatraemia does not cause episodic sweating. **B**: hPL-driven resistance is a second/third-trimester phenomenon and would raise glucose. **E**: possible in type 1 but does not explain episodic neuroglycopenia.

6. B · Use effective contraception while improving control (target HbA1c below 6.5%); stop glimepiride and lisinopril, switching to insulin/metformin and nifedipine or methyldopa; folic acid 5 mg daily from 3 months pre-conception

B is correct. High periconception HbA1c raises anomaly and miscarriage risk; aim for **HbA1c below 48 mmol/mol (6.5%)** before conception, **stop ACE inhibitors** (fetotoxic) and non-metformin oral agents, and give **folic acid 5 mg**. **A**: organogenesis happens before control improves. **C**: lisinopril and glimepiride should be stopped and 0.4 mg folate is insufficient. **D**: stopping treatment worsens hyperglycaemia. **E**: aspirin is started from 12 weeks of pregnancy and does not address control.

7. E · Start an IV insulin–dextrose (variable-rate or GKI) infusion and check glucose hourly aiming for 4–7 mmol/L

E is correct. Intrapartum target is **4–7 mmol/L** with **hourly** checks; hyperglycaemia in labour causes fetal hyperinsulinaemia, acidosis and neonatal hypoglycaemia. **A**: subcutaneous premixed insulin in labour is unpredictable (she may not eat, delivery timing uncertain). **C** misses the window. **D**: glucose of 9.4 is not an indication for CS. **B**: dextrose would worsen hyperglycaemia.

8. C · Stop immediately after delivery and check glucose before discharge

C is correct. Insulin resistance disappears with the placenta, so **GDM treatment is stopped** at delivery; check glucose before discharge to detect persisting hyperglycaemia, then test at **6–13 weeks**. **A**, **B**, **E**: unnecessary continuation (possible overt diabetes is detected by testing, not by continuing drugs blindly). **D**: insulin would risk hypoglycaemia.

9. A · IV 0.9% saline, fixed-rate IV insulin 0.1 unit/kg/hour, potassium replacement in fluids, left lateral tilt and fetal monitoring as the mother is stabilised

A is correct. This is **DKA**. Treat the mother: fluids, **fixed-rate insulin 0.1 unit/kg/h**, **K^+ 40 mmol/L** (K^+ 4.6 is in the 3.5–5.5 range), dextrose when glucose falls below 14, and treat the trigger. Fetal heart changes reflect maternal acidosis and usually **resolve with correction**. **B**: operating on an acidotic, hypovolaemic mother risks maternal death and delivers an acidotic preterm baby. **C** is inadequate and dangerous. **D**: bicarbonate is not routinely used. **E** is inadequate for DKA.

10. D · Assess fetal wellbeing urgently (CTG, growth scan, liquor, umbilical artery Doppler)

D is correct. A significant fall in insulin requirement in late pregnancy can signal **placental insufficiency** (falling placental hormone output), so the fetus must be assessed. **A**: dose reduction may be needed for safety, but reassurance alone misses the warning sign. **C**: diabetes does not resolve. **B** treats the number not the cause. **E**: she has known diabetes.

11. B · Nifedipine for tocolysis if indicated, dexamethasone 6 mg IM 12-hourly for 4 doses, with increased insulin and 2–4-hourly glucose monitoring

B is correct. Steroids are **not withheld** in diabetes, but they raise glucose for several days, so **increase insulin** and monitor closely. **Nifedipine** is the preferred tocolytic. **A**: beta-agonists cause hyperglycaemia and DKA. **C**: denies the baby proven benefit. **D**: stopping insulin during steroid therapy risks severe hyperglycaemia/DKA. **E**: $MgSO_4$ is used for fetal neuroprotection below 32 weeks, not as a first-line tocolytic, and steroids should not be withheld.

12. E · IV 10% dextrose 2 mL/kg bolus then a maintenance dextrose infusion, admit to the neonatal unit and monitor glucose

E is correct. A **symptomatic** hypoglycaemic baby needs **IV 10% dextrose 2 mL/kg**, then a maintenance infusion, with continued breastfeeding when possible. **A**: water provides no glucose. **C**: 50% dextrose is hyperosmolar and can cause rebound hypoglycaemia (hyperinsulinaemic baby) and vein damage. **D**: jitteriness with low glucose is a warning sign. **B**: discharging a symptomatic baby is unsafe.

13. C · Offer elective caesarean section

C is correct. In a diabetic woman, an EFW of **4,500 g or more** justifies offering **elective CS** (ACOG) because of the high risk of shoulder dystocia and brachial plexus injury. **A**: induction for suspected macrosomia is reasonable at lower EFW but here CS should be offered. **B**: delaying increases stillbirth and dystocia risk. **D**: instrumental delivery of a macrosomic baby increases shoulder dystocia. **E**: irrelevant (cephalic presentation assumed).

14. A · High risk of type 2 diabetes: lifestyle advice, diabetes-prevention support and annual testing

A is correct. NICE: fasting below 6.0 = low risk; **6.0–6.9 = high risk** (prediabetes-range), managed with lifestyle intervention and annual testing; 7.0 or more = likely type 2 diabetes. **B** under-reacts. **C** and **D** over-diagnose. **E**: she needs annual screening regardless of future pregnancy, plus an early OGTT if she conceives again.

15. D · Refer to the nearest HC IV/hospital for a 75 g OGTT at 24–28 weeks and explain why

D is correct. She has two risk factors (**glycosuria 2+** and **previous macrosomia**) and needs a **diagnostic OGTT** now (she is within 24–28 weeks). **A**: glycosuria can be physiological, but 2+ plus risk factors demands testing. **B** and **C**: treating without diagnosis is inappropriate and risky. **E**: carbohydrate restriction causes ketosis and is not advised.

SELF-TEST

Clinical cases

Work through each step before reading the model answer.

Case 1: GDM found in a woman with a previous big baby

Florence, 33, G4P3+0 at 26 weeks, BMI 34, attends Mengo ANC. Her last baby weighed 4.3 kg and 'got stuck at the shoulders'. Her mother has diabetes. Urine dipstick shows glucose 1+.

1. What further history do you take, and why?

> Previous pregnancies: birth weights, shoulder dystocia details, stillbirths, neonatal hypoglycaemia/admission (clues to undiagnosed GDM). Symptoms: polyuria, polydipsia, recurrent candidiasis/UTIs. Medicines (e.g. TLD, steroids). Diet, activity, occupation, ability to afford strips. Fetal movements.

2. What investigation do you arrange and how should it be done?

> **75 g OGTT at 24–28 weeks** (now): 8–14-hour overnight fast, fasting venous glucose, **75 g anhydrous glucose in 250–300 mL water over 5 minutes**, rest, glucose at **1 and 2 hours**. Result: fasting 5.8, 1-hour 10.9, 2-hour 9.1 mmol/L, meeting WHO 2013 criteria for **GDM** (every value is abnormal; none reach overt-diabetes thresholds).

3. How do you manage her initially?

> Education (cause, risks, plan). **Diet**: regular meals, low-GI carbohydrate in moderate portions with beans, greens, fish; no sugary drinks. **Walk 10–15 minutes after meals**. **Self-monitoring**: fasting and 1-hour post-meal. **Targets**: fasting below 5.3, 1-hour below 7.8. Review in 1–2 weeks. Growth scan.

4. Two weeks later fasting readings are 5.6–6.0 and post-meal 7.0–8.5. What next?

> Start **metformin 500 mg with the evening meal**, increasing by 500 mg every few days to a week up to about **2 g/day** in divided doses with meals. If fasting remains above 5.3 on the maximum dose, add **bedtime isophane insulin (about 0.1–0.2 units/kg)** and titrate by 2 units every 2–3 days.

5. How do you plan delivery?

> Growth scans at 28–32 and 36 weeks. If EFW is normal, plan **induction at about 38–39 weeks** (unit protocol). If EFW is **4.5 kg or more**, offer **CS**. Labour: hospital delivery, **hourly glucose, target 4–7 mmol/L**, insulin–dextrose infusion if above range, **senior present for possible shoulder dystocia**, active third stage (PPH risk).

6. What happens after delivery?

> **Stop metformin/insulin.** Check her glucose before discharge. Baby: breastfeed within 1 hour, **glucose at 2–4 hours** and before feeds for 24 hours; look for birth injury, jaundice, respiratory distress. **Fasting glucose at 6–13 weeks** (or OGTT), **annual screening for life**, early OGTT in the next pregnancy, weight and lifestyle advice, contraception.

Take-away: Risk factors mean OGTT; treat in steps (diet, metformin, insulin) to NICE targets; plan delivery around fetal size; stop treatment after birth and screen for life.

Case 2: Type 1 diabetes in labour

Brenda, 22, P0, with type 1 diabetes since age 12 on basal-bolus human insulin, is admitted at 37+4 weeks to a regional referral hospital for induction. Her HbA1c at booking was 8.4% and a scan at 36 weeks showed AC above the 95th centile with polyhydramnios.

1. What are the key risks in this labour?

> Maternal: hypoglycaemia, hyperglycaemia/DKA (especially if nil by mouth or on dextrose without insulin), PPH (polyhydramnios, big baby), operative delivery. Fetal: **shoulder dystocia**, **intrapartum hypoxia/stillbirth**, cord prolapse (polyhydramnios). Neonatal: **hypoglycaemia**, RDS, jaundice, polycythaemia, birth trauma.

2. How do you manage her glucose from the start of induction/labour?

> Usual bedtime basal insulin the night before (consider reducing by a quarter). From established labour (or the morning of induction if nil by mouth): **IV dextrose (5–10% with KCl) at about 100 mL/hour plus variable-rate IV insulin** (50 units soluble in 50 mL saline). **Hourly capillary glucose, target 4–7 mmol/L**. Never leave a type 1 woman without insulin. If no syringe driver: **GKI** (500 mL 10% dextrose + 10 units soluble insulin + 10 mmol KCl at 100 mL/hour).

3. Glucose falls to 3.4 mmol/L during labour. What do you do?

> Treat hypoglycaemia: if she can drink, 15–20 g fast sugar; if not, **IV 10% dextrose 150 mL** (or 20% dextrose 75 mL); reduce the insulin rate (type 1: do not stop completely for long; e.g. 0.5 unit/hour) and recheck in 15 minutes. Review the sliding scale.

4. What precautions for delivery?

> Continuous CTG. **Senior obstetrician in the second stage**, team briefed on **shoulder dystocia** (McRoberts, suprapubic pressure, internal manoeuvres). Avoid mid-cavity instrumental delivery for delay with suspected macrosomia; favour CS if progress arrests. Active management of third stage; oxytocin infusion ready. Neonatal team present.

5. What happens to her insulin after delivery?

> Insulin needs **fall abruptly** after placental delivery: reduce the infusion rate immediately, and when eating restart subcutaneous insulin at the **pre-pregnancy dose** (or about half the late-pregnancy dose), overlapping with the infusion by 30–60 minutes. Warn about **hypoglycaemia during breastfeeding**. Contraception and pre-conception counselling for the next pregnancy.

Take-away: In type 1 diabetes, insulin and dextrose always travel together in labour, glucose is kept at 4–7 mmol/L hourly, and insulin must be cut sharply the moment the placenta is out.

Case 3: DKA at 30 weeks

Night call: Joan, 24, type 1 diabetes, 30 weeks, arrives at a general hospital with 2 days of vomiting, fever and loin pain. She stopped her insulin 'because she was not eating'. RR 32, HR 124, BP 94/60, temperature 38.6 °C, drowsy. Capillary glucose 14.8 mmol/L, urine ketones 3+, nitrites positive. Fetal heart 170 bpm.

1. What is the diagnosis and the likely trigger?

> **Diabetic ketoacidosis** (ketonuria 3+, Kussmaul-type breathing, dehydration, drowsiness) triggered by **acute pyelonephritis** (fever, loin pain, nitrites) and **omitted insulin**. Glucose is only moderately raised, typical of pregnancy. Fetal tachycardia reflects maternal fever and acidosis.

2. What are your first 30 minutes of actions?

> Call for help (obstetrician, physician, anaesthetist). ABCDE, oxygen if SpO_2 below 94%, **left lateral tilt**. Two large cannulae; bloods: glucose, ketones, **K^+**, creatinine, bicarbonate/blood gas, FBC, **blood culture**, malaria test; urine culture. **0.9% saline 500 mL rapid bolus** (she is tachycardic and borderline hypotensive; repeat if SBP stays below 90) then 1 L over 1 hour. **Fixed-rate IV insulin 0.1 unit/kg/hour**. **IV ceftriaxone 1–2 g** after cultures. Paracetamol 1 g for fever. Catheter for hourly urine output.

3. How do you manage potassium and glucose over the next hours?

> Check K^+: if **3.5–5.5 mmol/L add 40 mmol KCl per litre**; none if above 5.5; senior review if below 3.5. When glucose falls **below 14 mmol/L**, add **10% dextrose at 125 mL/hour** and **continue insulin** until ketones clear. Monitor glucose and ketones hourly, K^+ and bicarbonate at 1, 2, 4 hours then 4-hourly; fluid balance; watch for pulmonary oedema.

4. The CTG shows reduced variability with late decelerations. The house officer wants to do a CS. What do you advise?

> **Do not operate on an unstable, acidotic mother.** Fetal heart abnormalities in DKA are due to maternal acidosis and hypovolaemia and **usually resolve as she is corrected**. Continue resuscitation, reassess the CTG, give **dexamethasone** only after discussion (it will worsen glucose control). Deliver only if fetal compromise persists after maternal stabilisation.

5. How do you complete her care?

> When ketones are below 0.6 and she is eating, restart subcutaneous basal-bolus insulin, overlapping with the infusion. Complete antibiotics for pyelonephritis (10–14 days, oral cephalosporin when afebrile 24–48 hours). LMWH thromboprophylaxis. **Sick-day rules education**: never stop basal insulin, check ketones when unwell, seek help early. Increase ANC surveillance.

Take-away: In pregnancy DKA comes faster and at lower glucose; resuscitate the mother (fluids, insulin, potassium, dextrose, treat the trigger), and the fetus usually recovers with her.

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A study aid, not a protocol. Check doses against the Uganda Clinical Guidelines 2023, your unit's protocol and a senior before you treat a patient.